Linear Algebra and Its Applications

Chapter 5

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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5.1 Eigenvectors and eigenvalues

An eigenvector of A is a *nonzero* vector v for which $Av = \lambda v$ for some scalar λ. The scalar is the eigenvalue: the direction stays on the same line, while the vector is scaled, possibly reversed or sent to zero. Rearranging gives

$$(A - \lambda I)v = 0.$$

Thus the eigenspace E_λ is $\mathrm{Nul}(A - \lambda I)$, and λ is an eigenvalue exactly when this null space contains a nonzero vector. The eigenspace includes zero; zero itself is never an eigenvector. To find eigenvectors for a specified eigenvalue, reduce $A - \lambda I$, not A. Row reduction of A itself generally changes its eigenvalues.

Practice problem 5.1.P1: Test five without guessing an eigenvector

The supplied matrix gives

$$A - 5I = \begin{bmatrix} 1 & -3 & 1 \\ 3 & -5 & 5 \\ 2 & 2 & 1 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - 3R_1$, $R_3 \leftarrow R_3 - 2R_1$:

$$\begin{bmatrix} 1 & -3 & 1 \\ 0 & 4 & 2 \\ 0 & 8 & -1 \end{bmatrix}.$$

Then $R_3 \leftarrow R_3 - 2R_2$ gives last row $(0, 0, -5)$. There are three pivots and no free variables, so $(A - 5I)x = 0$ has only zero as a solution. Hence 5 is not an eigenvalue.

Practice problem 5.1.P2: A power acting on an eigenvector

If $Ax = \lambda x$, then $A^2x = A(\lambda x) = \lambda Ax = \lambda^2 x$. Applying A once more gives $A^3x = \lambda^2 Ax = \lambda^3 x$. Each step uses linearity to pull out a scalar. Repeating the argument proves $A^k x = \lambda^k x$ for every nonnegative integer k.

Practice problem 5.1.P3: Several independent vectors within one eigenspace

Yes, all four vectors are independent. Suppose $c_1b_1 + c_2b_2 + c_3b_3 + c_4b_4 = 0$. Apply $A - \lambda_3 I$:

$$c_1(\lambda_1 - \lambda_3)b_1 + c_2(\lambda_2 - \lambda_3)b_2 = 0.$$

The two vectors b_1, b_2 correspond to distinct eigenvalues and are independent by Theorem 2. Since the eigenvalue differences are nonzero, $c_1 = c_2 = 0$. The original relation now reduces to $c_3b_3 + c_4b_4 = 0$, and their assumed independence forces $c_3 = c_4 = 0$. This proof also handles the possibility that $c_3b_3 + c_4b_4$ is zero; zero must not be called an eigenvector.

Exercise 5.1.1: Test a proposed eigenvalue

What we must show. Decide whether $\lambda = 2$ has a nonzero eigenvector. Reduce $A - \lambda I$, shown here as B:

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (3)R_1$. Then

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

The reduced equation is $x_1 + 2x_2 = 0$, with x_2 free. Taking $x_2 = 1$ gives $v = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \neq 0$. Directly $Av = \begin{bmatrix} -4 \\ 2 \end{bmatrix} = 2v$.

Conclusion. Yes, 2 is an eigenvalue; all nonzero multiples of v are corresponding eigenvectors.

Exercise 5.1.2: Test a proposed eigenvalue

What we must show. Decide whether $\lambda = -3$ has a nonzero eigenvector. Reduce $A - \lambda I$, shown here as B:

$$B = \begin{bmatrix} 2 & 4 \\ 6 & 12 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{2}R_1, \quad R_2 \leftarrow R_2 - (6)R_1$. Then

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

The reduced equation is $x_1 + 2x_2 = 0$, with x_2 free. Taking $x_2 = 1$ gives $v = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \neq 0$. Directly $Av = \begin{bmatrix} 6 \\ -3 \end{bmatrix} = -3v$.

Conclusion. Yes, -3 is an eigenvalue; all nonzero multiples of v are corresponding eigenvectors.

Exercise 5.1.3: Check a specified vector directly

The candidate is $v = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} 1 & -1 \\ 6 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -6 \end{bmatrix}.$$

This equals $-2v$ in every coordinate. Therefore v is an eigenvector with eigenvalue $\boxed{-2}$.

Exercise 5.1.4: Check a specified vector directly

The candidate is $v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} 5 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 3 \end{bmatrix}.$$

This equals $3v$ in every coordinate. Therefore v is an eigenvector with eigenvalue $\boxed{3}$.

Exercise 5.1.5: Check a specified vector directly

The candidate is $v = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} -4 & 3 & 3 \\ 2 & -3 & -2 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -15 \\ 10 \\ -5 \end{bmatrix}.$$

This equals $-5v$ in every coordinate. Therefore v is an eigenvector with eigenvalue $\boxed{-5}$.

Exercise 5.1.6: Check a specified vector directly

The candidate is $v = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} 3 & 6 & 7 \\ 3 & 2 & 7 \\ 5 & 6 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 13 \\ 1 \end{bmatrix}.$$

The first coordinate would force $\lambda = 5$, but the second would then have to be -10, not 13. A single common multiplier must work in every coordinate, so v is not an eigenvector.

Exercise 5.1.7: Produce a nonzero vector in the shifted null space

Solve $(A - 4I)x = 0$; the coefficient reduction is

$$A - 4I = \begin{bmatrix} -1 & 0 & -1 \\ 2 & -1 & 1 \\ -3 & 4 & 1 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2$, $\quad R_1 \leftarrow \frac{1}{2}R_1$, $\quad R_2 \leftarrow R_2 - (-1)R_1$, $\quad R_3 \leftarrow R_3 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{5}{2} & \frac{5}{2} \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_2 \leftarrow \frac{2}{5}R_2$, $R_1 \leftarrow R_1 - (-\frac{1}{2})R_2$, $R_3 \leftarrow R_3 - (-\frac{1}{2})R_2$. Then

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The free variable gives, for example, $v = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$. This is nonzero, and substitution gives $Av = \begin{bmatrix} -4 \\ -4 \\ 4 \end{bmatrix} = 4v$.

Conclusion. Yes. The specified scalar is an eigenvalue and v is a corresponding eigenvector.

Exercise 5.1.8: Produce a nonzero vector in the shifted null space

Solve $(A - 1I)x = 0$; the coefficient reduction is

$$A - 1I = \begin{bmatrix} 3 & -2 & 3 \\ 0 & -2 & 3 \\ -1 & 2 & -3 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{3}R_1$, $R_3 \leftarrow R_3 - (-1)R_1$. Then

$$\begin{bmatrix} 1 & -\frac{2}{3} & 1 \\ 0 & -2 & 3 \\ 0 & \frac{4}{3} & -2 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_2 \leftarrow \frac{3}{4}R_2$, $R_1 \leftarrow R_1 - (-\frac{2}{3})R_2$, $R_3 \leftarrow R_3 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\frac{3}{2} \\ 0 & 0 & 0 \end{bmatrix}.$$

The free variable gives, for example, $v = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}$. This is nonzero, and substitution gives $Av = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix} = 1v$.

Conclusion. Yes. The specified scalar is an eigenvalue and v is a corresponding eigenvector.

Exercise 5.1.9: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 1$.

$$A - 1I = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_1 = \text{Span}\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 1v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix}0 & 0\\2 & -2\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & -1\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_3 = \text{Span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.10: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix}-4 & 2\\3 & 1\end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = -5$.

$$A - (-5)I = \begin{bmatrix}1 & 2\\3 & 6\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & 2\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{-5} = \text{Span}\left\{\begin{bmatrix}-2\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = -5v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.11: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix}1 & -3\\-4 & 5\end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = -1$.

$$A - (-1)I = \begin{bmatrix}2 & -3\\-4 & 6\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & -\frac{3}{2}\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{-1} = \text{Span}\left\{\begin{bmatrix}3\\2\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = -1v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 7$.

$$A - 7I = \begin{bmatrix}-6 & -3\\-4 & -2\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & \frac{1}{2}\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{7} = \text{Span}\left\{\begin{bmatrix}-1\\2\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 7v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.12: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix}4 & 1\\3 & 6\end{bmatrix}$, solve $(A-\lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix}1 & 1\\3 & 3\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & 1\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{3} = \text{Span}\left\{\begin{bmatrix}-1\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 7$.

$$A - 7I = \begin{bmatrix}-3 & 1\\3 & -1\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & -\frac{1}{3}\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{7} = \text{Span}\left\{\begin{bmatrix}1\\3\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 7v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.13: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 1$.

$$A - 1I = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 1v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 2$.

$$A - 2I = \begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 2v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.14: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 4 & 0 & -1 \\ 3 & 0 & 3 \\ 2 & -2 & 5 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix} 1 & 0 & -1 \\ 3 & -3 & 3 \\ 2 & -2 & 2 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.15: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} -4 & 1 & 1 \\ 2 & -3 & 2 \\ 3 & 3 & -2 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = -5$.

$$A - (-5)I = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_{-5} = \operatorname{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent because they arise from distinct choices of a free coordinate; the pivot equations show that they span all solutions. Each satisfies $Av = -5v$ by direct substitution. Thus the displayed list is a basis, of dimension 2.

Exercise 5.1.16: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 5 & 0 & -1 & 0 \\ 1 & 3 & 0 & 0 \\ 2 & -1 & 3 & 0 \\ 4 & -2 & -2 & 4 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 4$.

$$A - 4I = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 2 & -1 & -1 & 0 \\ 4 & -2 & -2 & 0 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_4 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent because they arise from distinct choices of a free coordinate; the pivot equations show that they span all solutions. Each satisfies $Av = 4v$ by direct substitution. Thus the displayed list is a basis, of dimension 2.

Exercise 5.1.17: Read a triangular matrix's eigenvalues

The source matrix is upper triangular with diagonal entries $0, 3, -2$. For an upper triangular matrix, the eigenvalues are precisely its diagonal entries: $A - \lambda I$ loses a pivot exactly when one of these entries minus λ becomes zero.

Conclusion. The eigenvalues are $0, 3, -2$.

Exercise 5.1.18: The lower-triangular case

The source matrix is lower triangular with diagonal entries $5, 0, 3$. The triangular-matrix theorem therefore gives the eigenvalues directly.

Conclusion. The eigenvalues are $5, 0, 3$. Off-diagonal entries do not change this conclusion for a triangular matrix.

Exercise 5.1.19: A singular matrix has eigenvalue zero

All three columns are multiples of $(1, 1, 1)^T$, so they are dependent. Thus $Ax = 0$ has a nonzero solution and 0 is an eigenvalue. For example $(-2, 1, 0)^T$ gives a zero column combination, without needing elimination. The problem asks for one eigenvalue, not the complete list.

Exercise 5.1.20: Two independent vectors with coordinate sum zero

Each column of A is $(2,2,2)^T$. Therefore any input whose coordinates sum to zero is sent to zero. In particular

$$v_1 = (1,-1,0)^T, \qquad v_2 = (1,0,-1)^T$$

are eigenvectors for $\lambda = 0$. They are independent: in $av_1 + bv_2 = 0$, coordinate two forces $a = 0$ and coordinate three forces $b = 0$.

Conclusion. One eigenvalue is 0, with the two independent eigenvectors above.

Exercise 5.1.21: Do not omit the nonzero-vector requirement

(a) **False.** Taking $x = 0$ makes $Ax = \lambda x$ true for any scalar, even one that is not an eigenvalue. The definition requires $x \neq 0$.

(b) **True.** Noninvertibility is equivalent to $Ax = 0$ having a nonzero solution, precisely an eigenvector for eigenvalue zero.

(c) **True.** Rearranging $Ax = cx$ gives $(A - cI)x = 0$, with the same nonzero x required in both formulations.

(d) **True.** Compute Ax and check whether it equals one scalar multiple of the specified nonzero x. Finding the vector from scratch is a different task.

(e) **False.** Row operations on A need not preserve its eigenvalues. For example $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ has eigenvalues $2, 0$, but replacing row two by row two minus row one gives triangular $\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}$, with eigenvalues $1, 0$.

Exercise 5.1.22: Eigenspaces and repeated eigenvalues

(a) **False.** Zero solves the equality but is not an eigenvector. The condition $x \neq 0$ is missing.

(b) **False.** The independent vectors e_1, e_2 are both eigenvectors of I_2 with the same eigenvalue 1.

(c) **True.** A steady-state probability vector q is nonzero and satisfies $Mq = q$, so it is an eigenvector for eigenvalue 1.

(d) **False in general.** The matrix $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$ has eigenvalues $1, -1$, not the diagonal entry 0. The diagonal rule applies to triangular matrices.

(e) **True.** $E_\lambda = \mathrm{Nul}(A - \lambda I)$ by rearranging the eigenvector equation; the null space also includes zero.

Exercise 5.1.23: Bound the number of distinct eigenvalues without a polynomial

If an $n \times n$ matrix had p distinct eigenvalues, choose one nonzero eigenvector for each. Theorem 2 says those p vectors are independent. A list of vectors in $\mathbb{R}^n$ cannot be independent when $p > n$.

Hence $p \leq n$. In particular a 2×2 matrix has at most two distinct eigenvalues. This bounds distinct values, not the number of different eigenvectors, which can be infinite.

Exercise 5.1.24: An example with one distinct eigenvalue

Take $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I_2$. Every nonzero vector satisfies $Ax = 2x$. If $Ax = \lambda x$ with $x \neq 0$, then $(2 - \lambda)x = 0$ forces $\lambda = 2$. Thus the only distinct eigenvalue is 2, although there are infinitely many corresponding eigenvectors.

Exercise 5.1.25: Invert the eigenvalue as well as the matrix

Choose $x \neq 0$ with $Ax = \lambda x$. Since A is invertible, $\lambda \neq 0$: otherwise $Ax = 0$ would force $x = 0$. Multiply the eigenvector equation by A^{-1}: $x = \lambda A^{-1}x$, so $A^{-1}x = \lambda^{-1}x$ after division by the nonzero scalar. The vector x is still nonzero, proving that λ^{-1} is an eigenvalue of A^{-1} with the same eigenvector.

Exercise 5.1.26: A square-zero matrix can have only eigenvalue zero

If $Ax = \lambda x$ with $x \neq 0$, applying A again gives $0 = A^2x = \lambda^2 x$. A nonzero vector times a scalar is zero only when the scalar is zero, so $\lambda = 0$. Zero really is an eigenvalue: if $A = 0$, every nonzero vector works; if $A \neq 0$, choose y with $Ay \neq 0$ and set $x = Ay$. Then $x \neq 0$ but $Ax = A^2y = 0$.

Conclusion. The only eigenvalue is 0.

Exercise 5.1.27: Transpose preserves eigenvalues

The identity $A^T - \lambda I = (A - \lambda I)^T$ is the key. A matrix B is invertible if and only if B^T is invertible: if B^{-1} exists, $(B^{-1})^T$ is an inverse of B^T, and applying the same argument to B^T proves the converse. Therefore $A - \lambda I$ is singular exactly when $A^T - \lambda I$ is singular. Singularity of the shifted matrix is equivalent to the existence of a nonzero eigenvector.

Conclusion. A and A^T have the same eigenvalues, though their eigenvectors need not agree.

Exercise 5.1.28: Deduce the lower-triangular theorem by transposing

If A is lower triangular, A^T is upper triangular and has the same diagonal entries. The established upper-triangular case says that the eigenvalues of A^T are precisely these entries. Exercise 27 says A has exactly the same eigenvalues, completing the lower-triangular case without repeating an elimination argument.

Exercise 5.1.29: Equal row sums produce an obvious eigenvector

Let $\mathbf{1} = (1, \ldots, 1)^T \neq 0$. Entry i of $A\mathbf{1}$ is the sum of row i, which is s by hypothesis. Thus $A\mathbf{1} = (s, \ldots, s)^T = s\mathbf{1}$.

Conclusion. s is an eigenvalue, with eigenvector $\mathbf{1}$.

Exercise 5.1.30: Equal column sums use the transpose

The column sums of A are the row sums of A^T. Exercise 29 therefore makes s an eigenvalue of A^T. By Exercise 27 it is also an eigenvalue of A. The all-ones vector is an eigenvector of A^T here; it need not itself be an eigenvector of A, so the transpose step is substantive.

Exercise 5.1.31: Read eigenvalues from a reflection

Every nonzero vector on the reflecting line L is fixed, so $\lambda = 1$ is an eigenvalue and $E_1 = L$. A vector perpendicular to L is reversed, so -1 is the other eigenvalue, with $E_{-1} = L^{\perp}$. No vector outside L can be fixed: decompose it into its along-line and perpendicular parts; the reflection reverses the latter, so equality requires that part to be zero.

Exercise 5.1.32: Read a fixed eigenspace from a rotation

Every vector on the rotation axis L is fixed, so 1 is an eigenvalue. For a nontrivial rotation, the perpendicular-plane part of a nonzero vector is rotated and cannot remain unchanged, giving $E_1 = L$. If the angle is a whole multiple of 2π, the transformation is instead the identity and $E_1 = \mathbb{R}^3$. This qualification handles the degenerate zero-angle rotation, which the wording does not exclude.

Exercise 5.1.33: Combine two eigenvector solutions of a recurrence

(a) Shift the index in the definition.

The next term is $x_{k+1} = c_1\lambda^{k+1}u + c_2\mu^{k+1}v$.

(b) Apply A to the stated expression.

Using $Au = \lambda u$, $Av = \mu v$, and linearity,

$$Ax_k = c_1\lambda^k Au + c_2\mu^k Av = c_1\lambda^{k+1}u + c_2\mu^{k+1}v = x_{k+1}.$$

Thus the sequence satisfies the difference equation at every nonnegative index. Zeroth powers are taken as 1, including for a zero eigenvalue, as required by the initial term.

Exercise 5.1.34: Try expanding the initial state in eigenvectors

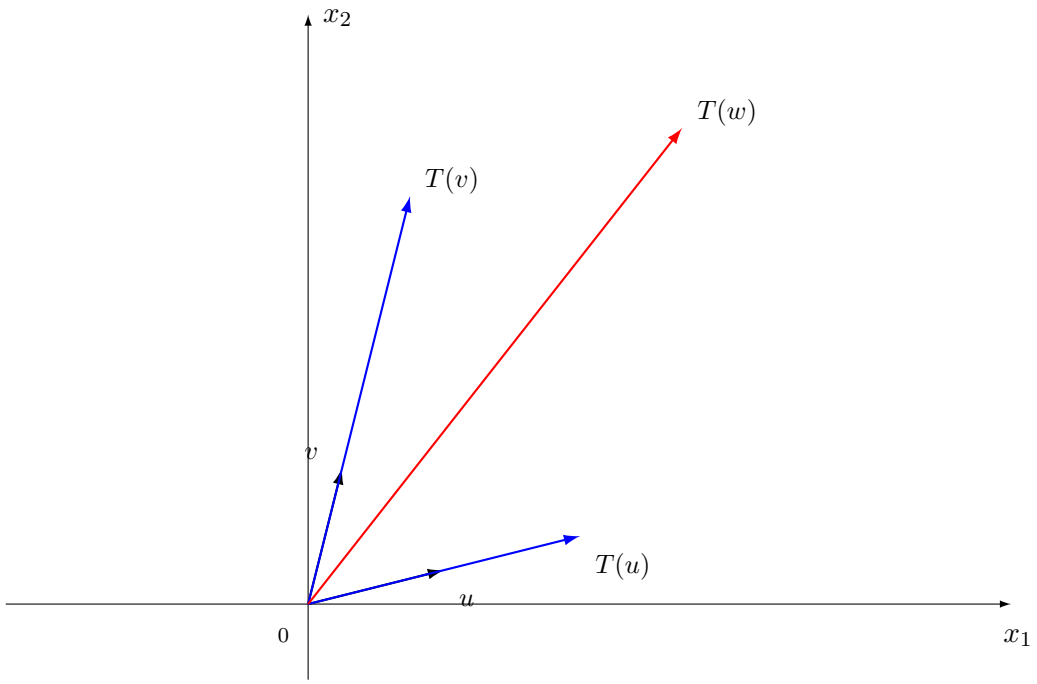
Find eigenvectors v_j of A and try solving $x_0 = \sum_j c_j v_j$ for coefficients c_j. If this is possible, set $x_k = \sum_j c_j \lambda_j^k v_j$, using the corresponding eigenvalues. At $k = 0$ it equals the prescribed state, and the calculation in Exercise 33 proves $Ax_k = x_{k+1}$ term by term. This method works whenever x_0 lies in the span of the available eigenvectors; in particular it works for every x_0 when there is an eigenvector basis. Not every matrix has such a basis, so this is a useful construction rather than an unconditional guarantee.

Exercise 5.1.35: Sketch the action on two eigendirections

The eigenvector equations give $T(u) = 2u$ and $T(v) = 3v$. Since $w = u + v$, linearity gives $T(w) = 2u + 3v$. Thus double the arrow along u, triple the arrow along v, and add those two new

arrows head to tail.

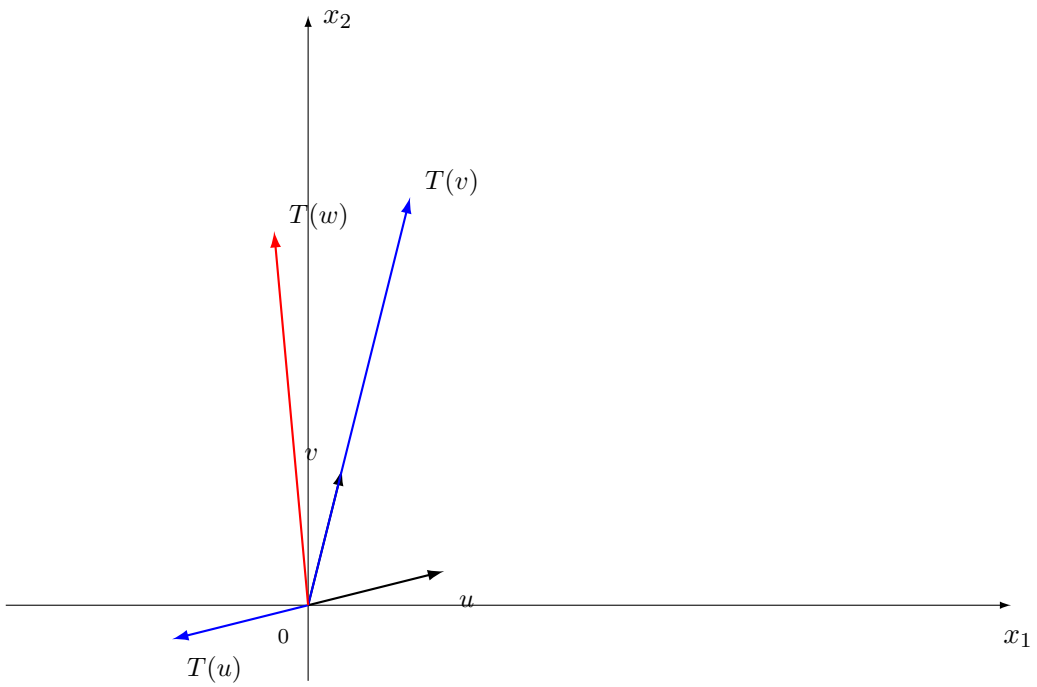
The black arrows reproduce the two source directions schematically; blue shows the separate images and red their sum. No numerical coordinates are assumed for the textbook's unlabeled vectors.



Exercise 5.1.36: Sketch the action on two eigendirections

The eigenvector equations give $T(u) = -1u$ and $T(v) = 3v$. Since $w = u + v$, linearity gives $T(w) = -1u + 3v$. Reverse the arrow u without changing its length, triple the arrow v, and add the resulting arrows head to tail.

The black arrows reproduce the two source directions schematically; blue shows the separate images and red their sum. No numerical coordinates are assumed for the textbook's unlabeled vectors.



Exercise 5.1.37: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A = \begin{bmatrix} 12 & 1 & 4 \\ 2 & 11 & 4 \\ 1 & 3 & 7 \end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A - \lambda I) = -(\lambda - 15)(\lambda - 10)(\lambda - 5).$$

Eigenvalue 5**.**

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_5 = \left\{ \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = 5v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 10**.**

$$\operatorname{rref}(A-(10)I)=\begin{bmatrix}1&0&3\\0&1&-2\\0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{10}=\left\{\begin{bmatrix}-3\\2\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=10v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 15**.**

$$\operatorname{rref}(A-(15)I)=\begin{bmatrix}1&0&-2\\0&1&-2\\0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{15}=\left\{\begin{bmatrix}2\\2\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=15v$ by exact substitution. The eigenspace dimension is 1.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

Exercise 5.1.38: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A=\begin{bmatrix}5&-2&2&-4\\7&-4&2&-4\\4&-4&2&0\\3&-1&1&-3\end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A-\lambda I)=(\lambda-2)\,(\lambda-1)\,(\lambda+1)\,(\lambda+2)\,.$$

Eigenvalue -2**.**

$$\operatorname{rref}(A-(-2)I)=\begin{bmatrix}1&0&0&0\\0&1&-1&0\\0&0&0&1\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-2}=\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=-2v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue -1**.**

$$\operatorname{rref}(A-(-1)I)=\begin{bmatrix}1&0&0&-1\\0&1&0&-1\\0&0&1&0\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-1}=\left\{\begin{bmatrix}1\\1\\0\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=-1v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 1**.**

$$\operatorname{rref}(A-(1)I)=\begin{bmatrix}1&0&0&-2\\0&1&0&-2\\0&0&1&0\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{1}=\left\{\begin{bmatrix}2\\2\\0\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=1v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 2**.**

$$\operatorname{rref}(A-(2)I)=\begin{bmatrix}1&0&0&-2\\0&1&0&-2\\0&0&1&-1\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_2=\left\{\begin{bmatrix}2\\2\\1\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=2v$ by exact substitution. The eigenspace dimension is 1.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

Exercise 5.1.39: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A=\begin{bmatrix}12&-90&30&30&30\\8&-49&15&15&15\\16&-52&12&0&20\\0&-30&10&22&10\\8&-41&15&15&7\end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A-\lambda I)=-(\lambda-12)^2(\lambda+4)(\lambda+8)^2.$$

Eigenvalue -8**.**

$$\operatorname{rref}(A-(-8)I)=\begin{bmatrix}1&0&0&-3&0\\0&1&0&-\frac{3}{2}&0\\0&0&1&-\frac{3}{2}&1\\0&0&0&0&0\\0&0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-8}=\left\{\begin{bmatrix}6\\3\\3\\2\\0\end{bmatrix},\ \begin{bmatrix}0\\0\\-1\\0\\1\end{bmatrix}\right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = -8v$ by exact substitution. The eigenspace dimension is 2.

Eigenvalue -4**.**

$$\operatorname{rref}(A - (-4)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-4} = \left\{ \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = -4v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 12**.**

$$\operatorname{rref}(A - (12)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{12} = \left\{ \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = 12v$ by exact substitution. The eigenspace dimension is 2.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

Exercise 5.1.40: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A = \begin{bmatrix} -23 & 57 & -9 & -15 & -59 \\ -10 & 12 & -10 & 2 & -22 \\ 11 & 5 & -3 & -19 & -15 \\ -27 & 31 & -27 & 25 & -37 \\ -5 & -15 & -5 & 1 & 31 \end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A - \lambda I) = -(\lambda - 42)^2 (\lambda + 14)^3.$$

Eigenvalue -14**.**

$$\operatorname{rref}(A - (-14)I) = \begin{bmatrix} 1 & 0 & 1 & 0 & -\frac{10}{3} \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 0 & 1 & -\frac{5}{3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-14} = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 10 \\ 6 \\ 0 \\ 5 \\ 3 \end{bmatrix} \right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = -14v$ by exact substitution. The eigenspace dimension is 2.

Eigenvalue 42**.**

$$\operatorname{rref}(A - (42)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & -\frac{1}{5} & \frac{1}{5} \\ 0 & 0 & 1 & \frac{2}{5} & \frac{3}{5} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{42} = \left\{ \begin{bmatrix} 0 \\ 1 \\ -2 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -1 \\ -3 \\ 0 \\ 5 \end{bmatrix} \right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = 42v$ by exact substitution. The eigenspace dimension is 2.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

5.2 The characteristic equation

We use the book's convention $p(\lambda) = \det(A - \lambda I)$. Some software instead reports $\det(\lambda I - A)$; they differ by $(-1)^n$ and have the same zeros. An eigenvalue is a zero of p, and its algebraic multiplicity is its multiplicity as a polynomial root. That multiplicity need not equal the dimension of its eigenspace. For a triangular matrix, the characteristic determinant is the product of the shifted diagonal entries. For a general matrix, use the full determinant rather than multiplying the original diagonal entries.

Practice problem 5.2.P1: A characteristic equation with no real roots

For the given matrix,

$$p(\lambda) = \det \begin{bmatrix} 1-\lambda & -4 \\ 4 & 2-\lambda \end{bmatrix} = (1-\lambda)(2-\lambda) + 16 = \lambda^2 - 3\lambda + 18.$$

Its discriminant is $9 - 72 = -63 < 0$, so the characteristic equation has no real root and A has no real eigenvalue. If complex scalars are allowed, its eigenvalues are $(3 \pm i\sqrt{63})/2$.

Exercise 5.2.1: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (2-\lambda)(2-\lambda) - (7)(7) = \lambda^2 - 4\lambda - 45.$$

Factoring gives $p(\lambda) = (\lambda - 9)\,(\lambda + 5)$. Its real roots, hence the eigenvalues, are $-5,\ 9$. Each listed root has multiplicity one.

Exercise 5.2.2: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} -4 & -1 \\ 6 & 1 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (-\lambda - 4)(1-\lambda) - (-1)(6) = \lambda^2 + 3\lambda + 2.$$

Factoring gives $p(\lambda) = (\lambda + 1)\,(\lambda + 2)$. Its real roots, hence the eigenvalues, are $-2,\ -1$. Each listed root has multiplicity one.

Exercise 5.2.3: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} -4 & 2 \\ 6 & 7 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (-\lambda - 4)(7-\lambda) - (2)(6) = \lambda^2 - 3\lambda - 40.$$

Factoring gives $p(\lambda) = (\lambda - 8)\,(\lambda + 5)$. Its real roots, hence the eigenvalues, are $-5,\ 8$. Each listed root has multiplicity one.

Exercise 5.2.4: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 8 & 2 \\ 3 & 3 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (8 - \lambda)(3 - \lambda) - (2)(3) = \lambda^2 - 11\lambda + 18.$$

Factoring gives $p(\lambda) = (\lambda - 9)(\lambda - 2)$. Its real roots, hence the eigenvalues, are $2,\ 9$. Each listed root has multiplicity one.

Exercise 5.2.5: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 8 & 4 \\ 4 & 8 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (8 - \lambda)(8 - \lambda) - (4)(4) = \lambda^2 - 16\lambda + 48.$$

Factoring gives $p(\lambda) = (\lambda - 12)(\lambda - 4)$. Its real roots, hence the eigenvalues, are $4,\ 12$. Each listed root has multiplicity one.

Exercise 5.2.6: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 9 & -2 \\ 2 & 5 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (9 - \lambda)(5 - \lambda) - (-2)(2) = \lambda^2 - 14\lambda + 49.$$

Factoring gives $p(\lambda) = (\lambda - 7)^2$. Its real roots, hence the eigenvalues, are 7. The root 7 has multiplicity two.

Exercise 5.2.7: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 5 & 3 \\ -4 & 4 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (5 - \lambda)(4 - \lambda) - (3)(-4) = \lambda^2 - 9\lambda + 32.$$

The discriminant is $(-9)^2 - 4(32) = -47 < 0$. Consequently there are no real eigenvalues. The two complex roots, if desired, are $(9 \pm i\sqrt{47})/2$.

Exercise 5.2.8: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} -4 & 3 \\ 2 & 1 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (-\lambda - 4)(1 - \lambda) - (3)(2) = \lambda^2 + 3\lambda - 10.$$

Factoring gives $p(\lambda) = (\lambda - 2)(\lambda + 5)$. Its real roots, hence the eigenvalues, are $-5,\ 2$. Each listed root has multiplicity one.

Exercise 5.2.9: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A-\lambda I=\begin{bmatrix}4-\lambda & 0 & -1\\ 0 & 4-\lambda & -1\\ 1 & 0 & 2-\lambda\end{bmatrix}.$$

Expand down column two; only its middle entry is nonzero. Thus

$$p(\lambda)=(4-\lambda)[(4-\lambda)(2-\lambda)+1].$$

Multiplying the factors gives

$$\boxed{p(\lambda)=-\lambda^3+10\lambda^2-33\lambda+36}.$$

A useful factored check is $p(\lambda)=-(\lambda-4)(\lambda-3)^2$. Expanding this expression reproduces the same cubic and, at $\lambda=0$, the original determinant.

Exercise 5.2.10: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A-\lambda I=\begin{bmatrix}3-\lambda & 1 & 1\\ 0 & 5-\lambda & 0\\ -2 & 0 & 7-\lambda\end{bmatrix}.$$

Expand across row two; only its middle entry is nonzero. Thus

$$p(\lambda)=(5-\lambda)[(3-\lambda)(7-\lambda)+2].$$

Multiplying the factors gives

$$\boxed{p(\lambda)=-\lambda^3+15\lambda^2-73\lambda+115}.$$

A useful factored check is $p(\lambda)=-(\lambda-5)(\lambda^2-10\lambda+23)$. Expanding this expression reproduces the same cubic and, at $\lambda=0$, the original determinant.

Exercise 5.2.11: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A-\lambda I=\begin{bmatrix}3-\lambda & 0 & 0\\ 2 & 1-\lambda & 4\\ 1 & 0 & 4-\lambda\end{bmatrix}.$$

Expand down column two; only its middle entry is nonzero, and the remaining minor is triangular. Thus

$$p(\lambda)=(1-\lambda)(3-\lambda)(4-\lambda).$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 8\lambda^2 - 19\lambda + 12}.$$

A useful factored check is $p(\lambda) = -(\lambda - 4)(\lambda - 3)(\lambda - 1)$. Expanding this expression reproduces the same cubic and, at $\lambda = 0$, the original determinant.

Exercise 5.2.12: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A - \lambda I = \begin{bmatrix} -\lambda - 1 & 0 & 2 \\ 3 & 1 - \lambda & 0 \\ 0 & 1 & 2 - \lambda \end{bmatrix}.$$

Expand across row one. The middle term is zero; the last cofactor contributes $2(3) = 6$. Thus

$$p(\lambda) = (-1 - \lambda)(1 - \lambda)(2 - \lambda) + 6.$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 2\lambda^2 + \lambda + 4}.$$

Factoring is not required here. As checks, the leading term is $-\lambda^3$ under our convention, and substituting $\lambda = 0$ gives $\det A = 4$.

Exercise 5.2.13: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A - \lambda I = \begin{bmatrix} 6 - \lambda & -2 & 0 \\ -2 & 9 - \lambda & 0 \\ 5 & 8 & 3 - \lambda \end{bmatrix}.$$

Expand down column three; only its last entry is nonzero. Thus

$$p(\lambda) = (3 - \lambda)[(6 - \lambda)(9 - \lambda) - 4].$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 18\lambda^2 - 95\lambda + 150}.$$

A useful factored check is $p(\lambda) = -(\lambda - 10)(\lambda - 5)(\lambda - 3)$. Expanding this expression reproduces the same cubic and, at $\lambda = 0$, the original determinant.

Exercise 5.2.14: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 0 & -1 \\ -1 & -\lambda & 4 \\ 0 & 2 & 3-\lambda \end{bmatrix}.$$

Expand across row one. The first minor gives $(-\lambda)(3-\lambda) - 8$, and the last cofactor contributes $(-1)(-2) = 2$. Thus

$$p(\lambda) = (4-\lambda)[(-\lambda)(3-\lambda) - 8] + 2.$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 7\lambda^2 - 4\lambda - 30}.$$

A useful factored check is $p(\lambda) = -(\lambda - 5)(\lambda^2 - 2\lambda - 6)$. Expanding this expression reproduces the same cubic and, at $\lambda = 0$, the original determinant.

Exercise 5.2.15: Read triangular eigenvalues with repetition

This matrix is upper triangular. Therefore $p(\lambda)$ is the product of its diagonal entries minus λ:

$$p(\lambda) = (\lambda - 5)^2 (\lambda - 3)(\lambda - 2).$$

Listing the diagonal entries, with each occurrence retained, gives the requested eigenvalue list 2, 3, 5, 5. Repeated entries are repeated eigenvalues in this list, not extra distinct values.

Exercise 5.2.16: Read triangular eigenvalues with repetition

This matrix is lower triangular. Therefore $p(\lambda)$ is the product of its diagonal entries minus λ:

$$p(\lambda) = (\lambda - 6)(\lambda - 3)(\lambda - 2)(\lambda + 5).$$

Listing the diagonal entries, with each occurrence retained, gives the requested eigenvalue list -5, 2, 3, 6. Repeated entries are repeated eigenvalues in this list, not extra distinct values.

Exercise 5.2.17: Read triangular eigenvalues with repetition

This matrix is lower triangular. Therefore $p(\lambda)$ is the product of its diagonal entries minus λ:

$$p(\lambda) = -\lambda (\lambda - 3)^2 (\lambda - 1)^2.$$

Listing the diagonal entries, with each occurrence retained, gives the requested eigenvalue list 0, 1, 1, 3, 3. Repeated entries are repeated eigenvalues in this list, not extra distinct values.

Exercise 5.2.18: Force a two-dimensional eigenspace for four

Subtract $4I$ to obtain

$$A - 4I = \begin{bmatrix} 0 & 2 & 3 & 3 \\ 0 & -2 & h & 3 \\ 0 & 0 & 0 & 14 \\ 0 & 0 & 0 & -2 \end{bmatrix}.$$

The last row of $(A - 4I)x = 0$ forces $x_4 = 0$. The first row then gives $2x_2 + 3x_3 = 0$, so $x_2 = -3x_3/2$. Substituting into the second gives $(h+3)x_3 = 0$. The coordinate x_1 is always free. For the eigenspace to have a second free coordinate, we need $h + 3 = 0$. At $h = -3$ its basis can be chosen as

$$(1, 0, 0, 0)^T, \qquad (0, -3, 2, 0)^T.$$

For any other h, $x_3 = x_2 = x_4 = 0$, leaving only one free coordinate.

Conclusion. $\boxed{h = -3}$.

Exercise 5.2.19: The constant coefficient is the determinant

The supplied factorization is $p(\lambda) = \prod_{j=1}^{n}(\lambda_j - \lambda)$, with eigenvalues repeated according to multiplicity. Substitute $\lambda = 0$:

$$\det A = \det(A - 0I) = p(0) = \lambda_1 \cdots \lambda_n.$$

No sign change is needed under this convention; using $\det(\lambda I - A)$ would introduce the corresponding constant-term sign. The same evaluation works over complex numbers when nonreal roots are present.

Exercise 5.2.20: Transpose preserves the whole characteristic polynomial

Since $I^T = I$ and transposition commutes with subtraction, $A^T - \lambda I = (A - \lambda I)^T$. The determinant of a transpose equals the original determinant. Hence

$$\det(A^T - \lambda I) = \det((A - \lambda I)^T) = \det(A - \lambda I)$$

for every λ. Thus the polynomials are identical, so the roots agree with their algebraic multiplicities, not merely as distinct sets.

Exercise 5.2.21: Determinants and the sign of a root

(a) **False.** For $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\det A = -1$, but the product of the diagonal entries is 0. The diagonal-product rule is valid for triangular matrices.

(b) **False.** A row interchange negates the determinant; scaling a row by c multiplies it by c. Only row replacement preserves it unchanged in general.

(c) **True.** This is the determinant product law for square matrices.

(d) **False.** A factor $\lambda + 5$ vanishes at $\lambda = -5$, not 5. For example $A = [-5]$ has that factor up to the harmless leading sign, but has only eigenvalue -5.

Exercise 5.2.22: Volume, transpose, multiplicity, and row operations

(a) **False as stated.** The geometric volume is $|\det A|$. The determinant can be negative because it also records orientation.

(b) **False.** $\det A^T = \det A$, with no factor -1. The identity matrix gives 1, not -1.

(c) **True.** This is the definition of algebraic multiplicity.

(d) **False.** Start with $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, whose eigenvalues are $2, 0$. Replacing row two by row two minus row one gives $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, whose eigenvalues are $1, 0$. Row replacement preserves the determinant, not every eigenvalue.

Exercise 5.2.23: Interchanging two factors gives a similar matrix

Assume $A = QR$ with Q invertible. Then

$$Q^{-1}AQ = Q^{-1}(QR)Q = (Q^{-1}Q)RQ = RQ = A_1.$$

Thus $A_1 = Q^{-1}AQ$, precisely the definition of similarity. Only Q needs to be invertible; R need not be. This is why interchanging the factors in the QR iteration preserves eigenvalues.

Exercise 5.2.24: Similarity preserves determinants

If $B = P^{-1}AP$ with P invertible, the product law gives

$$\det B = \det(P^{-1}) \det A \det P.$$

Since $\det(P^{-1}) \det P = \det I = 1$, the two outer factors cancel and $\det B = \det A$. This proof also covers singular A: only P, not A, is required to have an inverse.

Exercise 5.2.25: Resolve a probability vector into two eigenvectors

(a) Find a second eigendirection.

The stated $v_1 = (3/7, 4/7)^T$ satisfies $Av_1 = v_1$. For $v_2 = (1, -1)^T$, multiplication gives $Av_2 = (0.3, -0.3)^T = 0.3v_2$. The vectors correspond to distinct eigenvalues 1 and 0.3, so they are independent and form a basis of $\mathbb{R}^2$.

(b) Expand the initial vector.

$$x_0 - v_1 = (1/2 - 3/7, 1/2 - 4/7)^T = (1/14, -1/14)^T = \tfrac{1}{14}v_2.$$

Thus $x_0 = v_1 + (1/14)v_2$.

(c) Use the eigenvalue powers.

Direct multiplication gives $x_1 = (0.45, 0.55)^T$ and $x_2 = (0.435, 0.565)^T$. For every $k \geq 0$,

$$x_k = A^k x_0 = v_1 + \frac{1}{14}(0.3)^k v_2.$$

Because $|0.3| < 1$, the scalar factor tends to zero, so $x_k \to v_1$. This proves convergence, rather than merely inferring it from two similar iterates.

Exercise 5.2.26: Derive the two-by-two determinant from elimination

Case 1: $a \neq 0$**.**

Replace row two by row two minus (c/a) times row one. This operation does not change the determinant, and it produces $\begin{bmatrix} a & b \\ 0 & d - cb/a \end{bmatrix}$. The triangular formula gives $a(d - cb/a) = ad - bc$.

Case 2: $a = 0$**.**

If also $c = 0$, the first column is zero, so the determinant is zero, agreeing with $ad - bc$. If $c \neq 0$, interchange rows to produce $\begin{bmatrix} c & d \\ 0 & b \end{bmatrix}$. Its determinant is cb, and one interchange reverses the sign, so the original determinant is $-cb = ad - bc$. These cases include all possible values of a, c.

Exercise 5.2.27: Three eigendirections determine a steady-state limit

(a) Check the three given images.

With the matrix in the exercise, direct multiplication gives

$$Av_1 = (0.3, 0.6, 0.1)^T = v_1,$$

$$Av_2 = (0.5, -1.5, 1)^T = \tfrac{1}{2}v_2, \qquad Av_3 = (-0.2, 0, 0.2)^T = \tfrac{1}{5}v_3.$$

All three vectors are nonzero, so these really are eigenvector equations.

(b) Use both a basis and the coordinate sum.

The eigenvalues $1, 1/2, 1/5$ are distinct, so the three eigenvectors are independent and form a basis of $\mathbb{R}^3$. Thus any x_0 has unique coefficients c_j in $x_0 = c_1v_1 + c_2v_2 + c_3v_3$. For $w = (1, 1, 1)^T$, the probability hypothesis gives $w^T x_0 = 1$. But $w^T v_1 = 1$, $w^T v_2 = 0$, and $w^T v_3 = 0$. Applying w^T to the expansion therefore yields $1 = c_1$.

(c) Let the contracting components vanish.

$$x_k = A^k x_0 = v_1 + c_2(1/2)^k v_2 + c_3(1/5)^k v_3.$$

The two powers tend to zero, so $x_k \to v_1$. The two coefficients may depend on the initial distribution, but the limit does not, because its coefficient was fixed by total probability one.

Exercise 5.2.28: Reproducible random examples and the transpose distinction

To make the experiment reproducible, draw upper-triangular entries uniformly from the integers $-3, \ldots, 4$ with pseudorandom seed 52028, setting entries below the diagonal to zero. Successive draws of sizes four and five give the following valid random integer examples. The triangular choice makes exact checks of every multiplicity transparent.

$$A_4 = \begin{bmatrix} -1 & 0 & 0 & 2 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Exact calculation gives the same polynomial for A_4 and A_4^T:

$$p(\lambda) = (\lambda - 2)(\lambda - 1)^2(\lambda + 1).$$

Its eigenvalues, with multiplicity, are $-1,\ 1,\ 1,\ 2$. However e_1 is an eigenvector of A_4 with eigenvalue -1, whereas $A_4^T e_1 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 2 \end{bmatrix}$ is not a multiple of e_1. Thus their eigenvectors are not the same.

$$A_5 = \begin{bmatrix} 1 & 3 & 2 & 3 & 0 \\ 0 & -3 & -3 & 4 & -3 \\ 0 & 0 & 4 & -3 & 4 \\ 0 & 0 & 0 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Exact calculation gives the same polynomial for A_5 and A_5^T:

$$p(\lambda) = -\lambda(\lambda - 4)^2(\lambda - 1)(\lambda + 3).$$

Its eigenvalues, with multiplicity, are $-3,\ 0,\ 1,\ 4,\ 4$. However e_1 is an eigenvector of A_5 with eigenvalue 1, whereas $A_5^T e_1 = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 3 \\ 0 \end{bmatrix}$ is not a multiple of e_1. Thus their eigenvectors are not the same.

The equality of characteristic polynomials is guaranteed in general by Exercise 20; equal eigenspaces are not guaranteed. If a particular matrix is symmetric, its transpose is identical and its eigenspaces do agree, so one must not claim they always differ.

Exercise 5.2.29: Compare a determinant with pivot and eigenvalue products

One reproducible choice of a random integer matrix is

$$A = \begin{bmatrix} 2 & 3 & -3 & -2 \\ 5 & -3 & 0 & 3 \\ -3 & -4 & 0 & 2 \\ 1 & 1 & 4 & 1 \end{bmatrix}.$$

It was selected from pseudorandom entries in $-4, \dots, 5$ (seed 52029) with nonzero pivots and real eigenvalues, so every requested quantity can be displayed in real decimal form.

(a) Eliminate without row scaling.

First use $R_2 \leftarrow R_2 - (5/2)R_1$, $R_3 \leftarrow R_3 + (3/2)R_1$, and $R_4 \leftarrow R_4 - (1/2)R_1$. Continue replacing rows to clear entries below the second and third pivots. No swaps or row scalings are needed. The resulting echelon matrix is

$$U = \begin{bmatrix} 2 & 3 & -3 & -2 \\ 0 & -21/2 & 15/2 & 8 \\ 0 & 0 & -29/7 & -13/21 \\ 0 & 0 & 0 & 74/87 \end{bmatrix}.$$

Row replacement preserves the determinant, so

$$\det A = 2(-21/2)(-29/7)(74/87) = 74.$$

(b) Compute and multiply the eigenvalues.

Exact computation gives $p(\lambda) = \lambda^4 - 40\lambda^2 - 22\lambda + 74$. Its four real roots, computed with guard digits, are approximately

$$-5.8358727568, \quad -1.7435586500, \quad 1.1271321671, \quad 6.4522992397.$$

Their product using unrounded roots is 74 to numerical precision, as also follows exactly from the polynomial's constant coefficient.

(c) Give the requested four-decimal report and comparison.

The pivots are $2.0000, -10.5000, -4.1429, 0.8506$; the eigenvalues are $-5.8359, -1.7436, 1.1271, 6.4523$. The direct matrix determinant is exactly 74, agreeing with both products before rounding. Products formed only from the four-place display may differ slightly because the displayed numbers have been rounded.

Exercise 5.2.30: A small entry change can destroy two real eigenvalues

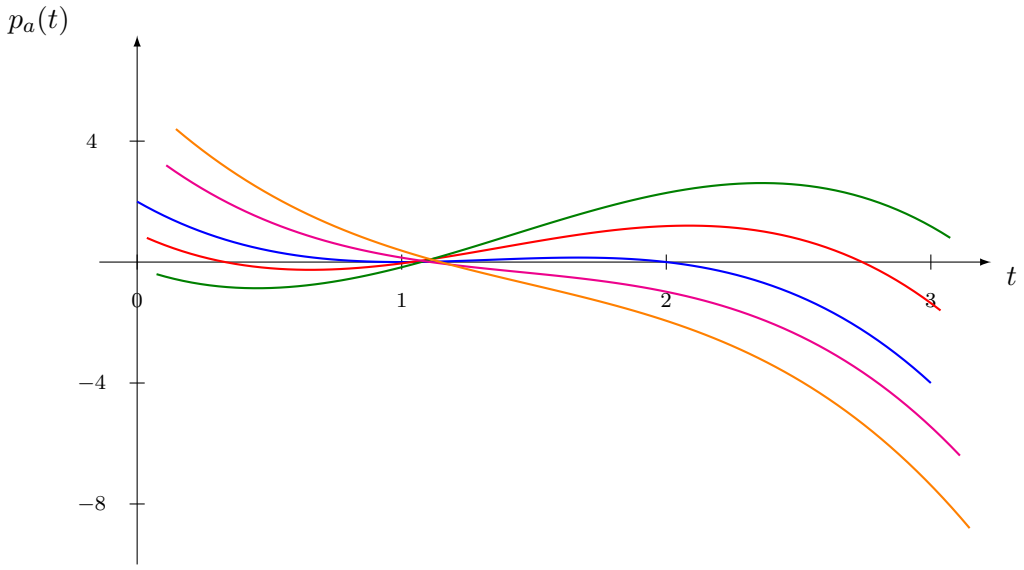
For the stated matrix, expansion of the shifted determinant gives

$$p_a(t) = -t^3 + 4t^2 + (379 - 12a)t + (12a - 382) = -(t-1)(t^2 - 3t + 12a - 382).$$

Thus 1 is always an eigenvalue, and the other two are $(3 \pm \sqrt{1537 - 48a})/2$. Substituting each requested value gives

a	$p_a(t)$	eigenvalues (approximately)
32	$-t^3 + 4t^2 - 5t + 2$	$1, 1, 2$
31.9	$-t^3 + 4t^2 - 3.8t + 0.8$	$0.295841, 1, 2.704159$
31.8	$-t^3 + 4t^2 - 2.6t - 0.4$	$-0.127882, 1, 3.127882$
32.1	$-t^3 + 4t^2 - 6.2t + 3.2$	$1, 1.5 \pm 0.974679i$
32.2	$-t^3 + 4t^2 - 7.4t + 4.4$	$1, 1.5 \pm 1.466288i$

The complex entries in the last two rows are the two nonreal eigenvalues; the only real eigenvalue in those cases is 1. Here are all five graphs on the requested common interval $0 \le t \le 3$.



$a = 32$, $a = 31.9$, $a = 31.8$, $a = 32.1$, $a = 32.2$.

Every graph passes through $(1, 0)$. At $a = 32$, the root 1 is double, so its graph touches the axis there and also crosses at 2. For $a = 31.9$, the three real crossings separate. For $a = 31.8$, two real roots move outside the plotted window; the graph within $[0, 3]$ shows only the crossing at 1, even though all three roots remain real. For $a = 32.1, 32.2$, the quadratic discriminant is negative, so those two roots are genuinely nonreal, not merely off screen. The coalescence threshold for that pair is $a = 1537/48 \approx 32.020833$.

Conclusion. The exact factorization explains the changes. A graph on a limited window alone cannot distinguish a root outside the window from a nonreal root; the discriminant and computed values supply that distinction.

5.3 Diagonalization

In this section diagonalization is over $\mathbb{R}$: only real eigenvectors can form the required basis of $\mathbb{R}^n$. Diagonalization is a change to an eigenvector coordinate system. If the columns of an invertible matrix P are eigenvectors and the corresponding eigenvalues are placed in the same order on the diagonal of D, then $AP = PD$ and $A = PDP^{-1}$. The requirement is *enough independent eigenvectors*, not merely enough vectors or roots of the characteristic polynomial.

The algebraic multiplicity of an eigenvalue is its number of appearances as a characteristic-polynomial root. Its geometric multiplicity is the dimension of its eigenspace. These are different notions. A repeated root must be checked by solving a homogeneous system; it neither guarantees nor forbids diagonalization.

Practice problem 5.3.P1: A high power from two eigenvectors

The characteristic polynomial is $(t-2)(t-1)$. Solving the two homogeneous systems gives eigenvectors $(3,2)$ for 2 and $(1,1)$ for 1. Thus

$$P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}.$$

In $(PDP^{-1})^8$, every adjacent $P^{-1}P$ cancels to I, giving $A^8 = PD^8P^{-1}$. Since $D^8 = \operatorname{diag}(256, 1)$,

$$A^8 = \begin{bmatrix} 766 & -765 \\ 510 & -509 \end{bmatrix}.$$

This result was also checked by eightfold exact matrix multiplication.

Practice problem 5.3.P2: Use the supplied eigenvectors

Multiplying by $A = \left[\begin{smallmatrix} -3 & 12 \\ -2 & 7 \end{smallmatrix}\right]$ gives $A(3,1)^T = (3,1)^T$ and $A(2,1)^T = 3(2,1)^T$. The eigenvalues are therefore 1 and 3, in that order. Take $P = \left[\begin{smallmatrix} 3 & 2 \\ 1 & 1 \end{smallmatrix}\right]$ and $D = \operatorname{diag}(1,3)$. Since $\det P = 1 \neq 0$ and $AP = PD$, we obtain $A = PDP^{-1}$.

Practice problem 5.3.P3: Count eigenspace dimensions

Yes. The eigenspace for 3 supplies two independent vectors. The distinct eigenvalues 5 and -2 each supply at least one nonzero eigenvector. Combining bases from distinct eigenspaces gives independent vectors, hence at least $2 + 1 + 1 = 4$ independent vectors in $\mathbb{R}^4$. There cannot be more than four. Thus they form an eigenvector basis and the matrix is diagonalizable.

Exercise 5.3.1: Cancel the inner change-of-basis factors

What we must show. Compute A^4 from the given $A = PDP^{-1}$. Associativity gives $(PDP^{-1})^4 = PD(P^{-1}P)D(P^{-1}P)D(P^{-1}P)DP^{-1} = PD^4P^{-1}$. Here

$$P = \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix}, \quad D^4 = \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix}.$$

Multiplying these three displayed matrices gives

$$A^4 = \begin{bmatrix} 226 & -525 \\ 90 & -209 \end{bmatrix}.$$

The calculation involves taking fourth powers of the diagonal entries of D, not taking fourth powers entry by entry in A.

Exercise 5.3.2: Cancel the inner change-of-basis factors

What we must show. Compute A^4 from the given $A = PDP^{-1}$. Associativity gives $(PDP^{-1})^4 = PD(P^{-1}P)D(P^{-1}P)D(P^{-1}P)DP^{-1} = PD^4P^{-1}$. Here

$$P = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \quad D^4 = \begin{bmatrix} 1 & 0 \\ 0 & 81 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}.$$

Multiplying these three displayed matrices gives

$$A^4 = \begin{bmatrix} 321 & -160 \\ 480 & -239 \end{bmatrix}.$$

The calculation involves taking fourth powers of the diagonal entries of D, not taking fourth powers entry by entry in A.

Exercise 5.3.3: An arbitrary power of a parameterized triangular matrix

Use the supplied factorization with $P = \left[\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right]$, $D = \operatorname{diag}(a, b)$, and $P^{-1} = \left[\begin{smallmatrix} 1 & 0 \\ -2 & 1 \end{smallmatrix}\right]$. For every positive integer k,

$$A^k = PD^kP^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} a^k & 0 \\ 0 & b^k \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} a^k & 0 \\ 2(a^k - b^k) & b^k \end{bmatrix}.$$

There is no division by $a - b$, so this remains true when $a = b$ or one parameter is zero.

Exercise 5.3.4: An arbitrary power with negative eigenvalues

Put $\alpha = (-3)^k$ and $\beta = (-2)^k$. The supplied matrices give

$$A^k = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} -3\alpha + 4\beta & 6\alpha - 6\beta \\ -2\alpha + 2\beta & 4\alpha - 3\beta \end{bmatrix}.$$

The signs in $(-3)^k$ and $(-2)^k$ depend on the parity of k and must remain inside parentheses. Setting $k = 1$ recovers the original matrix, a useful quick check.

Exercise 5.3.5: Read eigenspaces from an existing factorization

Read matching columns and diagonal entries in the supplied factorization. If p_j is column j of P, then $Ap_j = d_{jj}p_j$. Here

$$P = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

For $\lambda = 2$, a basis is $\{\begin{bmatrix}-1\\1\\-1\end{bmatrix}\}$. For $\lambda = 3$, a basis is $\{\begin{bmatrix}1\\-1\\0\end{bmatrix}, \begin{bmatrix}0\\-1\\1\end{bmatrix}\}$. Because P is invertible, its columns are independent. All columns at a repeated diagonal value are needed; retaining only one of them would lose an eigenspace direction.

Exercise 5.3.6: Read eigenspaces from an existing factorization

Read matching columns and diagonal entries in the supplied factorization. If p_j is column j of P, then $Ap_j = d_{jj}p_j$. Here

$$P = \begin{bmatrix}3 & 0 & -1\\0 & 1 & -3\\1 & 0 & 0\end{bmatrix}, \qquad D = \begin{bmatrix}3 & 0 & 0\\0 & 4 & 0\\0 & 0 & 3\end{bmatrix}.$$

For $\lambda = 3$, a basis is $\{\begin{bmatrix}3\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\-3\\0\end{bmatrix}\}$. For $\lambda = 4$, a basis is $\{\begin{bmatrix}0\\1\\0\end{bmatrix}\}$. Because P is invertible, its columns are independent. All columns at a repeated diagonal value are needed; retaining only one of them would lose an eigenspace direction.

Exercise 5.3.7: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix}1 & 0\\6 & -1\end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-1)(t+1).$$

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-1)I) = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \operatorname{Span}\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & -\frac{1}{3} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (1)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The exact checks are $\det P = -1 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.8: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 3 & 2 \\ 0 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 3)^2.$$

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 1 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 2, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.9: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-3)^2.$$

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 2, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.10: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t+2).$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & -\frac{3}{4} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 3 \\ -1 & 4 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 \\ 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 7 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.11: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 2 & 1 & 2 \\ 3 & 3 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t+1)^2.$$

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-1)I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & -\frac{1}{3} \\ 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ 0 & -1 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 6 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.12: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 5)\,(t - 2)^2.$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (2)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \text{Span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}1 & 1 & 1\\-1 & 0 & 1\\0 & -1 & 1\end{bmatrix}, \qquad D = \begin{bmatrix}2 & 0 & 0\\0 & 2 & 0\\0 & 0 & 5\end{bmatrix}.$$

The exact checks are $\det P = 3 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.13: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix}2 & 2 & -1\\1 & 3 & -1\\-1 & -2 & 2\end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t-1)^2.$$

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (1)I) = \begin{bmatrix}1 & 2 & -1\\0 & 0 & 0\\0 & 0 & 0\end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \text{Span}\left\{\begin{bmatrix}2\\-1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = -4 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.14: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & 0 & -2 \\ 1 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 3)^2 (t - 2).$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 1. Solve $(A-\lambda I)x=0$; row reduction gives

$$\operatorname{rref}(A-(2)I)=\begin{bmatrix}1&1&0\\0&0&1\\0&0&0\end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2=\operatorname{Span}\left\{\begin{bmatrix}1\\-1\\0\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av=(2)v$ for each listed vector.

Find the eigenspace for $\lambda=3$.

The algebraic multiplicity is 2. Solve $(A-\lambda I)x=0$; row reduction gives

$$\operatorname{rref}(A-(3)I)=\begin{bmatrix}1&0&2\\0&0&0\\0&0&0\end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3=\operatorname{Span}\left\{\begin{bmatrix}0\\1\\0\end{bmatrix},\ \begin{bmatrix}2\\0\\-1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av=(3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1+2=3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P=\begin{bmatrix}1&0&2\\-1&1&0\\0&0&-1\end{bmatrix},\qquad D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&3\end{bmatrix}.$$

The exact checks are $\det P=-1\neq 0$ and $AP=PD$. Multiplying the latter by P^{-1} on the right proves $A=PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.15: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A=\begin{bmatrix}0&-1&-1\\1&2&1\\-1&-1&0\end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = t\,(t-1)^2\,.$$

Find the eigenspace for $\lambda = 0$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (0)I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_0 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (0)v$ for each listed vector.

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (1)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 2 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The exact checks are $\det P = 1 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.16: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 5 & -2 \\ 1 & 3 & 1 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = t\left(t^2 - 7t + 16\right).$$

Find the eigenspace for $\lambda = 0$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (0)I) = \begin{bmatrix} 1 & 0 & -11 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_0 = \operatorname{Span}\left\{ \begin{bmatrix} 11 \\ -4 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (0)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 3, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction. The remaining quadratic factor is $t^2 - 7t + 16$, whose discriminant is $49 - 64 = -15 < 0$. Thus its other eigenvalues are nonreal. They do not supply real eigenvectors for a basis of $\mathbb{R}^3$; complex diagonalization is a different question introduced in Section 5.5.

Exercise 5.3.17: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-2)^3 .$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 3. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} .$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} .$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (2)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 3, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.18: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & -2 & -2 \\ 3 & -3 & -2 \\ 2 & -2 & -2 \end{bmatrix} .$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = t\,(t+1)\,(t+2) .$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} .$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \text{Span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (-1)I) = \begin{bmatrix}1 & 0 & -1\\0 & 1 & -\frac{1}{2}\\0 & 0 & 0\end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \text{Span}\left\{\begin{bmatrix}2\\1\\2\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 0$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (0)I) = \begin{bmatrix}1 & -1 & 0\\0 & 0 & 1\\0 & 0 & 0\end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{0} = \text{Span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (0)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}1 & 2 & 1\\1 & 1 & 1\\1 & 2 & 0\end{bmatrix}, \qquad D = \begin{bmatrix}-2 & 0 & 0\\0 & -1 & 0\\0 & 0 & 0\end{bmatrix}.$$

The exact checks are $\det P = 1 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.19: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 5 & -3 & 0 & 9 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t-3)(t-2)^2.$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (2)v$ for each listed vector.

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & -\frac{3}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 + 1 = 4$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 & 3 & 1 \\ 1 & -2 & 2 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = -2 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.20: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 3)^2 (t - 2)^2.$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (2)v$ for each listed vector.

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This is smaller than 4, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.21: Diagonalization criteria and constructions

(a) False.

The condition must say that D is *diagonal*. Without that, every matrix has the factorization $A = IAI^{-1}$, including a nondiagonalizable Jordan block.

(b) True.

A basis consists of n independent vectors; the Diagonalization Theorem applies to that eigenvector basis.

(c) False.

Counting roots with multiplicities does not count independent eigenvectors. For instance $\left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ has characteristic polynomial $(t-1)^2$ but only the one-dimensional eigenspace $\text{Span}\{(1,0)\}$.

(d) False.

A diagonal matrix can have a zero diagonal entry. The zero matrix is diagonalizable but not invertible.

Exercise 5.3.22: Diagonalization criteria and constructions

(a) False.

The vectors must be linearly independent. One eigenvector and its nonzero scalar multiples give arbitrarily many eigenvectors but only one direction.

(b) False.

I_n is diagonalizable and has only the eigenvalue 1 when $n > 1$.

(c) True.

Taking column j of $AP = PD$ gives $Ap_j = d_{jj}p_j$. If $p_j \neq 0$, this is exactly an eigenvector equation; zero columns are excluded by the definition.

(d) False.

The Jordan block $\left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ has determinant 1 and is invertible, but its eigenspace has dimension 1, so it is not diagonalizable.

Exercise 5.3.23: Diagonalization criteria and constructions

Yes. Choose a basis of three vectors from the first eigenspace and a basis of two from the other. Eigenvectors from different eigenspaces combine independently, so the union has $3+2 = 5$ independent vectors in $\mathbb{R}^5$. It is an eigenvector basis, giving diagonalizability.

Exercise 5.3.24: Diagonalization criteria and constructions

No. The dimensions of the only two eigenspaces add to $1 + 1 = 2$, which is less than 3. Every eigenvector belongs to one of those spaces, so one cannot obtain a third independent eigenvector. Thus the Diagonalization Theorem fails.

Exercise 5.3.25: Diagonalization criteria and constructions

No: under these assumptions the matrix must be diagonalizable. The two known dimensions are 1 and 2. The third eigenspace has at least dimension 1, giving at least $1 + 2 + 1 = 4$ independent

eigenvectors from the three distinct eigenspaces. Since the ambient space has dimension 4, this is exactly a basis.

Exercise 5.3.26: Diagonalization criteria and constructions

Yes, failure is possible. The known dimensions add to $2 + 3 = 5$, and the third can have dimension 1 or 2. If it has dimension 1, the total is $6 < 7$. For a concrete example take

$$A = \operatorname{diag}(0, 0, 1, 1, 1, J), \qquad J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

The eigenspace dimensions for $0, 1, 2$ are $2, 3, 1$, respectively, so this 7×7 matrix is not diagonalizable. If the last block were $2I_2$, the third dimension would instead be 2 and the matrix would be diagonalizable. The supplied data do not decide which occurs.

Exercise 5.3.27: Diagonalization criteria and constructions

Write $A = PDP^{-1}$. Invertibility of A implies no diagonal entry of D is zero. Inverting the product in reverse order gives $A^{-1} = PD^{-1}P^{-1}$, where D^{-1} is again diagonal. Thus A^{-1} is diagonalizable, and it is invertible because its inverse is A. Equivalently, $Av = \lambda v$ implies $A^{-1}v = \lambda^{-1}v$ for the same eigenvector basis.

Exercise 5.3.28: Diagonalization criteria and constructions

The hypothesis means $A = PDP^{-1}$ with D diagonal. Transpose this equality:

$$A^T = (P^{-1})^T D P^T = QDQ^{-1}, \qquad Q = (P^{-1})^T.$$

The matrix Q is invertible, and its columns therefore form n independent eigenvectors of A^T. The transpose generally does not have the *same* individual eigenvectors; the point is that it has its own eigenvector basis.

Exercise 5.3.29: Diagonalization criteria and constructions

In Example 2, $A = \left[\begin{smallmatrix} 7 & 2 \\ -4 & 1 \end{smallmatrix}\right]$, with eigenvectors $(1, -1)$ for 5 and $(1, -2)$ for 3. The new $D_1 = \operatorname{diag}(3, 5)$ reverses the eigenvalue order, so reverse the columns as well:

$$P_1 = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}.$$

Its determinant is $1 \neq 0$, and multiplication verifies $AP_1 = P_1 D_1$. Thus $A = P_1 D_1 P_1^{-1}$. Reordering eigenvalues without reordering eigenvectors would generally be wrong.

Exercise 5.3.30: Diagonalization criteria and constructions

Keep $D = \operatorname{diag}(5, 3)$ from Example 2, but rescale its first eigenvector. For instance,

$$P_2 = \begin{bmatrix} 2 & 1 \\ -2 & -2 \end{bmatrix}$$

has columns $2(1,-1)$ and $(1,-2)$ and determinant $-2 \neq 0$. It differs from the displayed P, but $AP_2 = P_2D$ still holds, so $A = P_2DP_2^{-1}$. Any nonzero rescaling of a basis eigenvector gives another valid factorization with that eigenvalue order.

Exercise 5.3.31: Diagonalization criteria and constructions

Take $A = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$. Its determinant is 1, so it is invertible and certainly nonzero. Its only eigenvalue is 1, and $(A - I)x = 0$ says $x_2 = 0$, leaving only a one-dimensional eigenspace. Therefore it is not diagonalizable.

Exercise 5.3.32: Diagonalization criteria and constructions

Take $A = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right]$. It is nondiagonal and has determinant 0, so is not invertible. Its distinct eigenvalues are 1 and 0; corresponding independent eigenvectors are $(1,0)$ and $(-1,1)$. Thus with $P = \left[\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right]$ and $D = \operatorname{diag}(1,0)$, we have $AP = PD$ and $A = PDP^{-1}$.

Exercise 5.3.33: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 9 & -4 & -2 & -4 \\ -56 & 32 & -28 & 44 \\ -14 & -14 & 6 & -14 \\ 42 & -33 & 21 & -45 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-13)^2\,(t+12)^2\,.$$

Find the eigenspace for $\lambda = -12$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-12)I) = \begin{bmatrix} 1 & 0 & -\frac{2}{7} & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-12} = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 7 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-12)v$ for each listed vector.

Find the eigenspace for $\lambda = 13$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (13)I) = \begin{bmatrix} 1 & 0 & \frac{1}{2} & -\frac{1}{3} \\ 0 & 1 & 0 & \frac{4}{3} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{13} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -4 \\ 0 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (13)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 2 = 4$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 0 & 1 & 1 \\ 7 & 1 & 0 & -4 \\ 7 & 0 & -2 & 0 \\ 0 & -1 & 0 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} -12 & 0 & 0 & 0 \\ 0 & -12 & 0 & 0 \\ 0 & 0 & 13 & 0 \\ 0 & 0 & 0 & 13 \end{bmatrix}.$$

The exact checks are $\det P = 25 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

Exercise 5.3.34: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 4 & -9 & -7 & 8 & 2 \\ -7 & -9 & 0 & 7 & 14 \\ 5 & 10 & 5 & -5 & -10 \\ -2 & 3 & 7 & 0 & 4 \\ -3 & -13 & -7 & 10 & 11 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)^3 (t+2)^2 .$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 0 & 0 & \frac{2}{5} & -\frac{3}{5} \\ 0 & 1 & 0 & -\frac{7}{5} & -\frac{2}{5} \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} .$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \operatorname{Span} \left\{ \begin{bmatrix} 2 \\ -7 \\ 5 \\ -5 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ -5 \\ 0 \\ 5 \end{bmatrix} \right\} .$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 3. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & -2 & 1 & -2 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} .$$

Columns 3, 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span} \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} .$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 3. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 3 = 5$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 3 & 2 & 1 & 2 \\ -7 & 7 & -1 & -1 & 0 \\ 5 & -5 & 1 & 0 & 0 \\ -5 & 0 & 0 & -1 & 0 \\ 0 & 5 & 0 & 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 35 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

Exercise 5.3.35: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 13 & -12 & 9 & -15 & 9 \\ 6 & -5 & 9 & -15 & 9 \\ 6 & -12 & -5 & 6 & 9 \\ 6 & -12 & 9 & -8 & 9 \\ -6 & 12 & 12 & -6 & -2 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 7)^3 (t + 14)^2.$$

Find the eigenspace for $\lambda = -14$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-14)I) = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-14} = \text{Span}\left\{\begin{bmatrix}1\\1\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\\-1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-14)v$ for each listed vector.

Find the eigenspace for $\lambda = 7$.

The algebraic multiplicity is 3. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (7)I) = \begin{bmatrix}1 & -2 & 0 & -1 & \frac{3}{2}\\0 & 0 & 1 & -1 & 0\\0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0\end{bmatrix}.$$

Columns 2, 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_7 = \text{Span}\left\{\begin{bmatrix}2\\1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\0\\0\\-2\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 3. Direct multiplication checks $Av = (7)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 3 = 5$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}1 & 0 & 2 & 1 & 3\\1 & 0 & 1 & 0 & 0\\0 & 1 & 0 & 1 & 0\\1 & 0 & 0 & 1 & 0\\0 & -1 & 0 & 0 & -2\end{bmatrix}, \qquad D = \begin{bmatrix}-14 & 0 & 0 & 0 & 0\\0 & -14 & 0 & 0 & 0\\0 & 0 & 7 & 0 & 0\\0 & 0 & 0 & 7 & 0\\0 & 0 & 0 & 0 & 7\end{bmatrix}.$$

The exact checks are $\det P = -7 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

Exercise 5.3.36: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 24 & -6 & 2 & 6 & 2 \\ 72 & 51 & 9 & -99 & 9 \\ 0 & -63 & 15 & 63 & 63 \\ 72 & 15 & 9 & -63 & 9 \\ 0 & 63 & 21 & -63 & -27 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 36)^2 (t - 24) (t + 48)^2 .$$

Find the eigenspace for $\lambda = -48$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-48)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-48} = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-48)v$ for each listed vector.

Find the eigenspace for $\lambda = 24$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (24)I) = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{24} = \text{Span}\left\{\begin{bmatrix}1\\1\\0\\1\\0\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (24)v$ for each listed vector.

Find the eigenspace for $\lambda = 36$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (36)I) = \begin{bmatrix}1 & 0 & 0 & -1 & \frac{1}{3}\\0 & 1 & 0 & 0 & -1\\0 & 0 & 1 & -3 & 0\\0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0\end{bmatrix}.$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{36} = \text{Span}\left\{\begin{bmatrix}1\\0\\3\\1\\0\end{bmatrix}, \begin{bmatrix}1\\-3\\0\\0\\-3\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (36)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 + 2 = 5$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}0 & 0 & 1 & 1 & 1\\1 & 0 & 1 & 0 & -3\\0 & 1 & 0 & 3 & 0\\1 & 0 & 1 & 1 & 0\\0 & -1 & 0 & 0 & -3\end{bmatrix}, \qquad D = \begin{bmatrix}-48 & 0 & 0 & 0 & 0\\0 & -48 & 0 & 0 & 0\\0 & 0 & 24 & 0 & 0\\0 & 0 & 0 & 36 & 0\\0 & 0 & 0 & 0 & 36\end{bmatrix}.$$

The exact checks are $\det P = -12 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

5.4 Eigenvectors and linear transformations

The matrix of a linear map depends on its *ordered* domain and codomain bases. Column j is the coordinate vector of the image of domain basis vector j, written in the codomain basis. For $T(x) = Ax$ on $\mathbb{R}^n$, if P has the basis vectors as its columns, then $[T]_{\mathcal{B}} = P^{-1}AP$.

The relation $[T(x)]_{\mathcal{B}} = [T]_{\mathcal{B}}[x]_{\mathcal{B}}$ uses coordinates on both sides. A coordinate column is not automatically the same as the vector it represents; they coincide only when using the standard basis. Similarity changes coordinates without changing the underlying linear transformation.

Practice problem 5.4.P1: Recover a polynomial from its coordinate column

Relative to $(1, t, t^2)$, the polynomial $p = a_0 + a_1t + a_2t^2$ has column $(a_0, a_1, a_2)^T$. Multiply by the supplied matrix:

$$[T(p)]_{\mathcal{B}} = \begin{bmatrix} 3 & 4 & 0 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 3a_0 + 4a_1 \\ 5a_1 - a_2 \\ a_0 - 2a_1 + 7a_2 \end{bmatrix}.$$

Translate this column back into the polynomial $T(p) = (3a_0 + 4a_1) + (5a_1 - a_2)t + (a_0 - 2a_1 + 7a_2)t^2$.

Practice problem 5.4.P2: Reflexivity and transitivity of similarity

(a) Every matrix is similar to itself.

Choose $P = I$; then $P^{-1}AP = A$ and P is invertible.

(b) Compose two changes of basis.

If $B = P^{-1}AP$ and $C = Q^{-1}BQ$, substitute the first equation into the second:

$$C = Q^{-1}P^{-1}APQ = (PQ)^{-1}A(PQ).$$

The product PQ is invertible, so this is exactly the definition saying C is similar to A. The reversed order in $(PQ)^{-1} = Q^{-1}P^{-1}$ is essential.

Exercise 5.4.1: Construct a matrix from basis-vector images

The codomain coordinate vectors of $T(b_1)$, $T(b_2)$, $T(b_3)$ are $(3, -5)^T$, $(-1, 6)^T$, and $(0, 4)^T$. Put them in columns, in that order:

$$[T]_{\mathcal{D}\leftarrow\mathcal{B}} = \begin{bmatrix} 3 & -1 & 0 \\ -5 & 6 & 4 \end{bmatrix}.$$

It has two rows (the codomain dimension) and three columns (the domain dimension). Multiplication by a column $(x_1, x_2, x_3)^T$ produces the coordinates of $x_1T(b_1) + x_2T(b_2) + x_3T(b_3)$, explaining the column rule.

Exercise 5.4.2: Construct a matrix from basis-vector images

The domain basis is now $\mathcal{D}$ and the codomain basis is $\mathcal{B}$. The supplied images have coordinate columns $(3,-3)^T$ and $(-2,5)^T$ in $\mathcal{B}$. Therefore

$$[T]_{\mathcal{B}\leftarrow\mathcal{D}} = \begin{bmatrix} 3 & -2 \\ -3 & 5 \end{bmatrix}.$$

The arrow in the subscript records the direction of coordinates and helps avoid reversing the bases.

Exercise 5.4.3: Construct a matrix from basis-vector images

(a) Substitute the standard basis vectors one at a time.

The formula gives $T(e_1) = b_3$, $T(e_2) = -b_1 - 2b_2$, and $T(e_3) = 2b_1 + 3b_3$.

(b) Read their coefficients in $\mathcal{B}$.

These columns are $(0,0,1)^T$, $(-1,-2,0)^T$, and $(2,0,3)^T$.

(c) Put them in the original basis order.

Thus the matrix relative to the standard domain basis and codomain basis $\mathcal{B}$ is

$$\begin{bmatrix} 0 & -1 & 2 \\ 0 & -2 & 0 \\ 1 & 0 & 3 \end{bmatrix}.$$

Multiplying by $(x_1,x_2,x_3)^T$ recovers exactly $(2x_3 - x_2, -2x_2, x_1 + 3x_3)^T$, the coefficients in the given transformation.

Exercise 5.4.4: Construct a matrix from basis-vector images

The input column in $\mathcal{B}$ is $(x_1,x_2,x_3)^T$, while the output is already in standard coordinates of $\mathbb{R}^2$. Reading the coefficient of each input scalar gives

$$[T]_{\mathcal{E}\leftarrow\mathcal{B}} = \begin{bmatrix} 2 & -3 & 1 \\ -2 & 0 & 5 \end{bmatrix}.$$

In particular the zero in row 2, column 2 is needed because the second output component does not contain x_2.

Exercise 5.4.5: Multiplication of polynomials as a linear map

(a) Multiply the specified polynomial, combining like powers.

$$T(3 - 2t + t^2) = (t+3)(3 - 2t + t^2) = t^3 + t^2 - 3t + 9.$$

(b) Verify linearity for arbitrary inputs and scalars.

For polynomials p, q and scalars c, d, distributivity gives

$$T(cp + dq) = (t + 3)(cp + dq) = cT(p) + dT(q).$$

The degree is at most 3, so the image does belong to the stated codomain.

(c) Apply the map to each monomial in the domain basis.

$T(1) = t + 3$. $T(t) = t^2 + 3t$. $T(t^2) = t^3 + 3t^2$. Writing those coefficients in increasing powers of t gives the matrix

$$C = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

The column lengths reflect the codomain degree bound, even when one image has lower degree and hence trailing zero coefficients.

Exercise 5.4.6: Multiplication of polynomials as a linear map

(a) Multiply the specified polynomial, combining like powers.

$$T(3 - 2t + t^2) = (2t^2 + 1)(3 - 2t + t^2) = 2t^4 - 4t^3 + 7t^2 - 2t + 3.$$

(b) Verify linearity for arbitrary inputs and scalars.

For polynomials p, q and scalars c, d, distributivity gives

$$T(cp + dq) = (2t^2 + 1)(cp + dq) = cT(p) + dT(q).$$

The degree is at most 4, so the image does belong to the stated codomain.

(c) Apply the map to each monomial in the domain basis.

$T(1) = 2t^2 + 1$. $T(t) = 2t^3 + t$. $T(t^2) = 2t^4 + t^2$. Writing those coefficients in increasing powers of t gives the matrix

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

The column lengths reflect the codomain degree bound, even when one image has lower degree and hence trailing zero coefficients.

Exercise 5.4.7: Coordinates determine a linear transformation

Evaluate the formula at the three basis polynomials: $T(1) = 3 + 5t$, $T(t) = -2t + 4t^2$, $T(t^2) = t^2$. Their coefficient columns give

$$[T]_{\mathcal{B}} = \begin{bmatrix} 3 & 0 & 0 \\ 5 & -2 & 0 \\ 0 & 4 & 1 \end{bmatrix}.$$

As a check, multiplying this by $(a_0, a_1, a_2)^T$ yields $(3a_0, 5a_0 - 2a_1, 4a_1 + a_2)^T$, exactly the supplied coefficient rule.

Exercise 5.4.8: Coordinates determine a linear transformation

The input has coordinate vector $(4, -3, 0)^T$. Thus

$$[T(4b_1 - 3b_2)]_{\mathcal{B}} = \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & -2 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ -5 \end{bmatrix}.$$

Translate back to the vector space: $T(4b_1 - 3b_2) = 5b_2 - 5b_3$. The result is not the coordinate column itself unless these basis vectors happen to be standard coordinate vectors.

Exercise 5.4.9: Evaluation at three input values

(a) Evaluate the given polynomial.

For $p(t) = 5 + 3t$, the three values are $p(-1) = 2$, $p(0) = 5$, and $p(1) = 8$, so $T(p) = (2, 5, 8)^T$.

(b) Establish linearity.

For any fixed input s, $(cp + dq)(s) = cp(s) + dq(s)$. Applying this at $s = -1, 0, 1$ componentwise proves $T(cp + dq) = cT(p) + dT(q)$.

(c) Evaluate the three basis polynomials.

The image columns for $1, t, t^2$ are $(1, 1, 1)^T$, $(-1, 0, 1)^T$, and $(1, 0, 1)^T$. Therefore

$$[T] = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

For example this matrix times $(5, 3, 0)^T$ returns the value found in part (a).

Exercise 5.4.10: Evaluation of a cubic at four points

(a) Linearity holds componentwise.

Evaluation at any fixed input distributes over polynomial addition and scalar multiplication. Applying this at $-2, 3, 1, 0$ proves the vector-valued map is linear.

(b) Respect the given order of evaluation points.

Evaluate each of $1, t, t^2, t^3$, putting its four values in a column. The matrix is

$$[T] = \begin{bmatrix} 1 & -2 & 4 & -8 \\ 1 & 3 & 9 & 27 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

For a general coefficient column $(a_0, a_1, a_2, a_3)^T$, multiplication gives the polynomial's values at those four points. The row for input 0 is last, not first, because the output order was specified that way.

Exercise 5.4.11: Change both input and output coordinates

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix}, \qquad \det P = 1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 2 & 3 \\ -4 & -5 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} -2 & -2 \\ 0 & -1 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order.

Exercise 5.4.12: Change both input and output coordinates

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}, \qquad \det P = 1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} -2 & 2 \\ 0 & -4 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} -4 & 0 \\ 2 & -2 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order.

Exercise 5.4.13: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 0 & 1 \\ -3 & 4 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-3)(t-1).$$

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (1)v$ for each listed vector.

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & -\frac{1}{3} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}.$$

The exact checks are $\det P = 2 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.14: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 5)(t + 1).$$

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-1)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{5} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 \\ 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 2 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.15: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-2)(t+5).$$

Find the eigenspace for $\lambda = -5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-5)I) = \begin{bmatrix} 1 & \frac{1}{3} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-5} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-5)v$ for each listed vector.

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{2} = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (2)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 2 \\ -3 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -5 & 0 \\ 0 & 2 \end{bmatrix}.$$

The exact checks are $\det P = 7 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.16: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 4 & -2 \\ -1 & 5 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-6)(t-3).$$

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Find the eigenspace for $\lambda = 6$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (6)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_6 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (6)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix}.$$

The exact checks are $\det P = -3 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.17: An eigenvector without an eigenvector basis

(a) Verify the stated eigenvector and inspect multiplicity.

Multiplication gives $A(1,-1)^T = (3,-3)^T = 3(1,-1)^T$. But

$$\det(tI - A) = (t-3)^2, \qquad A - 3I = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}.$$

The homogeneous equation is $x_1 + x_2 = 0$, so its eigenspace has only dimension 1. It cannot supply a basis for $\mathbb{R}^2$; thus A is not diagonalizable.

(b) Use the specified basis anyway.

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \qquad \det P = 1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 3 & -2 \\ -3 & 5 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order. The resulting matrix is not diagonal. A valid change of basis exists even when no diagonalizing change of basis exists.

Exercise 5.4.18: Similarity and coordinate identities

The eigenvalues alone do not decide the question. A diagonal basis matrix exists precisely when A is diagonalizable, which here means the eigenspace for 5 has dimension 2. The eigenspace for -2, whose algebraic multiplicity is 1, has dimension 1.

For a positive example take $A = \operatorname{diag}(5,5,-2)$; the standard basis works. For a negative example take $A = \left[\begin{smallmatrix} 5 & 1 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -2 \end{smallmatrix}\right]$. It has the same eigenvalues, but its 5-eigenspace is only one-dimensional. Thus one must compute an eigenspace or equivalent rank information before answering for a particular A.

Exercise 5.4.19: Similarity and coordinate identities

Write $B = P^{-1}AP$ with P invertible. The product is invertible because all three factors are, and reversing the factors gives

$$B^{-1} = P^{-1}A^{-1}P.$$

Thus the same change of basis $Q = P$ exhibits similarity of the inverses. It is legitimate to invert this expression because invertibility was established first.

Exercise 5.4.20: Similarity and coordinate identities

If $B = P^{-1}AP$, then $B^2 = P^{-1}AP\,P^{-1}AP = P^{-1}A^2P$, since the middle $PP^{-1} = I$. This is the required similarity factorization for the squares. The same cancellation would prove the result for any positive integer power.

Exercise 5.4.21: Similarity and coordinate identities

Write $B = P^{-1}AP$ and $C = Q^{-1}AQ$. Solving the second identity for A gives $A = QCQ^{-1}$. Substitution yields

$$B = P^{-1}QCQ^{-1}P = R^{-1}CR, \qquad R = Q^{-1}P.$$

The matrix R is invertible as a product of invertible matrices, so B and C are similar. This argument is the symmetry-plus-transitivity property in explicit form.

Exercise 5.4.22: Similarity and coordinate identities

Suppose $A = PDP^{-1}$ with D diagonal and $B = Q^{-1}AQ$. Then

$$B = (Q^{-1}P)D(P^{-1}Q).$$

The last factor is $(Q^{-1}P)^{-1}$. Thus this is a diagonalization of B, with an invertible change-of-basis matrix $Q^{-1}P$.

Exercise 5.4.23: Similarity and coordinate identities

Set $y = P^{-1}x$. This vector is nonzero: if $y = 0$, multiplication by P would force $x = 0$, contradicting the eigenvector definition. Now

$$By = P^{-1}AP\,P^{-1}x = P^{-1}Ax = P^{-1}(\lambda x) = \lambda y.$$

Therefore y is an eigenvector of B for the same eigenvalue. The nonzero check is an essential part of the proof.

Exercise 5.4.24: Similarity and coordinate identities

For $B = P^{-1}AP$, first note that AP has the same column space as A. Every column of AP is a combination of columns of A, and $A = (AP)P^{-1}$ supplies the reverse inclusion. Thus $\operatorname{rank}(AP) = \operatorname{rank} A$. Next, multiplication on the left by P^{-1} is an invertible linear map on the codomain. It maps the column space of AP bijectively onto that of B, preserving its dimension. Consequently $\operatorname{rank} B = \operatorname{rank} A$.

Exercise 5.4.25: Similarity and coordinate identities

Use the supplied cyclic identity on two square factors, $F = P^{-1}A$ and $G = P$:

$$\operatorname{tr}(B) = \operatorname{tr}((P^{-1}A)P) = \operatorname{tr}(P(P^{-1}A)) = \operatorname{tr}(A).$$

Only associativity and $PP^{-1} = I$ are used; this does not incorrectly commute arbitrary matrix factors.

Exercise 5.4.26: Similarity and coordinate identities

If $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, Exercise 25 gives $\operatorname{tr} A = \operatorname{tr} D$. The latter is the sum of the diagonal entries, hence $\sum_{j=1}^{n} \lambda_j$. These diagonal entries are the eigenvalues counted with their algebraic multiplicities in a diagonalization, proving the assertion in the requested case.

Exercise 5.4.27: Similarity and coordinate identities

The identity map sends the jth domain basis vector b_j to the same vector b_j, now expressed in the standard codomain basis. Thus its columns are $b_1, \dots, b_n$:

$$[I]_{\mathcal{E} \leftarrow \mathcal{B}} = [b_1 \ \cdots \ b_n] = P_{\mathcal{B}}.$$

This is the change-of-coordinates matrix from $\mathcal{B}$ coordinates to standard coordinates, also called the basis matrix in Section 4.4. Its action is $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$.

Exercise 5.4.28: Similarity and coordinate identities

The identity sends b_j to b_j, whose codomain coordinate column is $[b_j]_{\mathcal{C}}$. Therefore

$$[I]_{\mathcal{C} \leftarrow \mathcal{B}} = \big[[b_1]_{\mathcal{C}} \ \cdots \ [b_n]_{\mathcal{C}}\big].$$

This is the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$ in Section 4.7. Its defining equation is $[x]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}}[x]_{\mathcal{B}}$; the vector x itself does not change.

Exercise 5.4.29: Similarity and coordinate identities

With the same basis on both sides, $[I(b_j)]_{\mathcal{B}} = [b_j]_{\mathcal{B}} = e_j$. The matrix with those columns is exactly I_n. Hence $[I]_{\mathcal{B}} = I_n$ for every ordered basis, not merely the standard one.

Exercise 5.4.30: An exact basis-matrix computation

The standard matrix is

$$A = \begin{bmatrix} 6 & -2 & -2 \\ 3 & 1 & -2 \\ 2 & -2 & 2 \end{bmatrix}.$$

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & -1 \\ 1 & 3 & 0 \end{bmatrix}, \qquad \det P = -1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 2 & 4 & -4 \\ 2 & 1 & -4 \\ 2 & 8 & 0 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3 & 0 \\ 0 & 1 & 4 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order. This computer-assisted calculation can be reproduced exactly by setting the displayed matrices in SymPy and evaluating

```
C = P.inv()*A*P
assert P*C == A*P
print(C)
```

Solving the augmented system instead of explicitly forming the inverse gives the same result.

Exercise 5.4.31: An exact basis-matrix computation

The standard matrix is

$$A = \begin{bmatrix} -7 & -48 & -16 \\ 1 & 14 & 6 \\ -3 & -45 & -19 \end{bmatrix}.$$

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} -3 & -2 & 3 \\ 1 & 1 & -1 \\ -3 & -3 & 0 \end{bmatrix}, \qquad \det P = 3 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 21 & 14 & 27 \\ -7 & -6 & -11 \\ 21 & 18 & 36 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} -7 & -2 & -6 \\ 0 & -4 & -6 \\ 0 & 0 & -1 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order. This computer-assisted calculation can be reproduced exactly by setting the displayed matrices in SymPy and evaluating

```
C = P.inv()*A*P
assert P*C == A*P
print(C)
```

Solving the augmented system instead of explicitly forming the inverse gives the same result.

Exercise 5.4.32: Find a diagonal coordinate representation

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} -6 & 4 & 0 & 9 \\ -3 & 0 & 1 & 6 \\ -1 & -2 & 1 & 0 \\ -4 & 4 & 0 & 7 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t-1)(t+2)^2.$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 0 & -1 & -\frac{3}{2} \\ 0 & 1 & -1 & \frac{3}{4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -3 \\ 0 \\ 4 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{7}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ -1 \\ -7 \\ 2 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \\ -1 \\ 2 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 + 1 = 4$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 6 & 2 & 2 \\ 1 & -3 & -1 & 1 \\ 1 & 0 & -7 & -1 \\ 0 & 4 & 2 & 2 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 84 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.The columns of the displayed P form the requested basis; the corresponding matrix of the transformation is D.

5.5 Complex eigenvalues

A real matrix can have nonreal eigenvalues when it acts on complex coordinate vectors. Complex arithmetic uses $i^2 = -1$ and conjugation $\overline{a+bi} = a-bi$. For a real matrix, conjugating $Av = \lambda v$ gives $A\overline{v} = \overline{\lambda}\,\overline{v}$, so nonreal eigenvalues occur in conjugate pairs.

The real matrix $C = \left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$ is a scaling followed by a rotation: $C = r\left[\begin{smallmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{smallmatrix}\right]$, where $r = \sqrt{a^2+b^2}$, $\cos\varphi = a/r$, and $\sin\varphi = b/r$. The sign of b matters for clockwise versus counterclockwise rotation. We use angles in $(-\pi, \pi]$.

Practice problem 5.5.P1: Verify the two complex eigenpairs directly

Multiplication gives

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}\begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} a+bi \\ b-ai \end{bmatrix} = (a+bi)\begin{bmatrix} 1 \\ -i \end{bmatrix}.$$

Likewise $A(1,i)^T = (a-bi)(1,i)^T$. Both vectors are nonzero, so these are genuine eigenpairs. They are conjugate to one another. If $b = 0$, the eigenvalues coincide at a; the verified equations still hold, and in fact every nonzero vector is an eigenvector of aI_2.

Exercise 5.5.1: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 4t + 5 = 0$, whose two roots are $2-i,\ 2+i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 2-i$, the first equation of $(A-\lambda I)x = 0$ is

$$(-1+i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1-i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2-i} = \mathrm{Span}_{\mathbb{C}}\left\{\begin{bmatrix} -2 \\ 1-i \end{bmatrix}\right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 2+i$, the first equation of $(A-\lambda I)x = 0$ is

$$(-1-i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1+i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2+i} = \mathrm{Span}_{\mathbb{C}}\left\{\begin{bmatrix} -2 \\ 1+i \end{bmatrix}\right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.2: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 3 & -3 \\ 3 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 6t + 18 = 0$, whose two roots are $3 - 3i,\ 3 + 3i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 3 - 3i$, the first equation of $(A - \lambda I)x = 0$ is

$$(3i)x_1 + (-3)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -3s$ and $x_2 = (-3i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3-3i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -3 \\ -3i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 3 + 3i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-3i)x_1 + (-3)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -3s$ and $x_2 = (3i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3+3i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -3 \\ 3i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.3: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 5 & 1 \\ -8 & 1 \end{bmatrix}.$$

The characteristic equation is $t^2 - 6t + 13 = 0$, whose two roots are $3 - 2i,\ 3 + 2i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 3 - 2i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 + 2i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (-2 - 2i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3-2i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ -2 - 2i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 3 + 2i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 - 2i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (-2 + 2i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3+2i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ -2 + 2i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.4: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 4t + 5 = 0$, whose two roots are $2 - i,\ 2 + i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 2 - i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 + i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1 - i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2-i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -2 \\ 1 - i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 2 + i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 - i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1 + i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2+i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -2 \\ 1 + i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. The printed matrix in Exercise 4 is identical to that in Exercise 1; the repeated answer reflects the actual supplied edition, rather than an altered matrix.

Exercise 5.5.5: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 5 \end{bmatrix}.$$

The characteristic equation is $t^2 - 8t + 17 = 0$, whose two roots are $4 - i,\ 4 + i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 4 - i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 + i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (1 - i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{4-i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ 1 - i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 4 + i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 - i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (1 + i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{4+i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ 1 + i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.6: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 7 & -5 \\ 1 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 10t + 26 = 0$, whose two roots are $5 - i,\ 5 + i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 5 - i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 + i)x_1 + (-5)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -5s$ and $x_2 = (-2 - i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{5-i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -5 \\ -2 - i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 5 + i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 - i)x_1 + (-5)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -5s$ and $x_2 = (-2 + i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{5+i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -5 \\ -2 + i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.7: Rotation angle and positive scale

The matrix has the canonical entries $a = \sqrt{3}$ and $b = 1$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = \sqrt{3} \pm (1)i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = \frac{\sqrt{3}}{2}$ and $\sin\varphi = b/r = \frac{1}{2}$. In the requested interval these give

$$\varphi = \frac{\pi}{6}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.8: Rotation angle and positive scale

The matrix has the canonical entries $a = 3$ and $b = -3\sqrt{3}$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 3 \pm (-3\sqrt{3})i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 6.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = \frac{1}{2}$ and $\sin\varphi = b/r = -\frac{\sqrt{3}}{2}$. In the requested interval these give

$$\varphi = -\frac{\pi}{3}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.9: Rotation angle and positive scale

The matrix has the canonical entries $a = 0$ and $b = -2$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 0 \pm (-2)i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = 0$ and $\sin\varphi = b/r = -1$. In the requested interval these give

$$\varphi = -\frac{\pi}{2}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.10: Rotation angle and positive scale

The matrix has the canonical entries $a = 0$ and $b = -\frac{1}{2}$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 0 \pm (-\frac{1}{2})i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = \frac{1}{2}.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = 0$ and $\sin\varphi = b/r = -1$. In the requested interval these give

$$\varphi = -\frac{\pi}{2}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.11: Rotation angle and positive scale

The matrix has the canonical entries $a = -\sqrt{3}$ and $b = -1$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = -\sqrt{3} \pm (-1)i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = -\frac{\sqrt{3}}{2}$ and $\sin\varphi = b/r = -\frac{1}{2}$. In the requested interval these give

$$\varphi = -\frac{5\pi}{6}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.12: Rotation angle and positive scale

The matrix has the canonical entries $a = 3$ and $b = \sqrt{3}$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 3 \pm (\sqrt{3})i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2\sqrt{3}.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = \frac{\sqrt{3}}{2}$ and $\sin\varphi = b/r = \frac{1}{2}$. In the requested interval these give

$$\varphi = \frac{\pi}{6}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.13: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}.$$

The characteristic roots are $2 \pm (1)i$. Choose the root $\lambda = a - bi = 2 - i$ with $b = 1 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -2 \\ 1 - i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -2 & 0 \\ 1 & -1 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Here $\det P = 2 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.14: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} 3 & -3 \\ 1 & 1 \end{bmatrix}.$$

The characteristic roots are $2 \pm (\sqrt{2})i$. Choose the root $\lambda = a - bi = 2 - \sqrt{2}i$ with $b = \sqrt{2} > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -3 \\ -1 - \sqrt{2}i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -3 \\ -1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -\sqrt{2} \end{bmatrix}.$$

Equate real and imaginary parts in $A(u+iw)=(a-bi)(u+iw)$. This yields $Au=au+bw$ and $Aw=-bu+aw$, so in the ordered real basis (u,w) the matrix is C. Consequently take

$$P=\begin{bmatrix}-3 & 0\\ -1 & -\sqrt{2}\end{bmatrix},\qquad C=\begin{bmatrix}2 & -\sqrt{2}\\ \sqrt{2} & 2\end{bmatrix}.$$

Here $\det P=3\sqrt{2}\neq 0$. Direct exact multiplication verifies $AP=PC$, which, after multiplying by P^{-1}, gives the required $A=PCP^{-1}$.

Exercise 5.5.15: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A=\begin{bmatrix}0 & 5\\ -2 & 2\end{bmatrix}.$$

The characteristic roots are $1\pm(3)i$. Choose the root $\lambda=a-bi=1-3i$ with $b=3>0$. Solving its first homogeneous equation as above gives the eigenvector

$$v=\begin{bmatrix}5\\ 1-3i\end{bmatrix}=u+iw,\qquad u=\begin{bmatrix}5\\ 1\end{bmatrix},\quad w=\begin{bmatrix}0\\ -3\end{bmatrix}.$$

Equate real and imaginary parts in $A(u+iw)=(a-bi)(u+iw)$. This yields $Au=au+bw$ and $Aw=-bu+aw$, so in the ordered real basis (u,w) the matrix is C. Consequently take

$$P=\begin{bmatrix}5 & 0\\ 1 & -3\end{bmatrix},\qquad C=\begin{bmatrix}1 & -3\\ 3 & 1\end{bmatrix}.$$

Here $\det P=-15\neq 0$. Direct exact multiplication verifies $AP=PC$, which, after multiplying by P^{-1}, gives the required $A=PCP^{-1}$.

Exercise 5.5.16: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A=\begin{bmatrix}4 & -2\\ 1 & 6\end{bmatrix}.$$

The characteristic roots are $5\pm(1)i$. Choose the root $\lambda=a-bi=5-i$ with $b=1>0$. Solving its first homogeneous equation as above gives the eigenvector

$$v=\begin{bmatrix}-2\\ 1-i\end{bmatrix}=u+iw,\qquad u=\begin{bmatrix}-2\\ 1\end{bmatrix},\quad w=\begin{bmatrix}0\\ -1\end{bmatrix}.$$

Equate real and imaginary parts in $A(u+iw)=(a-bi)(u+iw)$. This yields $Au=au+bw$ and $Aw=-bu+aw$, so in the ordered real basis (u,w) the matrix is C. Consequently take

$$P=\begin{bmatrix}-2 & 0\\ 1 & -1\end{bmatrix},\qquad C=\begin{bmatrix}5 & -1\\ 1 & 5\end{bmatrix}.$$

Here $\det P=2\neq 0$. Direct exact multiplication verifies $AP=PC$, which, after multiplying by P^{-1}, gives the required $A=PCP^{-1}$.

Exercise 5.5.17: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} -11 & -4 \\ 20 & 5 \end{bmatrix}.$$

The characteristic roots are $-3 \pm (4)i$. Choose the root $\lambda = a - bi = -3 - 4i$ with $b = 4 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -4 \\ 8 - 4i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -4 \\ 8 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -4 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -4 & 0 \\ 8 & -4 \end{bmatrix}, \qquad C = \begin{bmatrix} -3 & -4 \\ 4 & -3 \end{bmatrix}.$$

Here $\det P = 16 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.18: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} 3 & -5 \\ 2 & 5 \end{bmatrix}.$$

The characteristic roots are $4 \pm (3)i$. Choose the root $\lambda = a - bi = 4 - 3i$ with $b = 3 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -5 \\ 1 - 3i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -5 \\ 1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -3 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -5 & 0 \\ 1 & -3 \end{bmatrix}, \qquad C = \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}.$$

Here $\det P = 15 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.19: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} \frac{38}{25} & -\frac{7}{10} \\ \frac{14}{25} & \frac{2}{5} \end{bmatrix}.$$

The characteristic roots are $\frac{24}{25} \pm (\frac{7}{25})i$. Choose the root $\lambda = a - bi = \frac{24}{25} - \frac{7i}{25}$ with $b = \frac{7}{25} > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -\frac{7}{10} \\ -\frac{14}{25} - \frac{7i}{25} \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -\frac{7}{10} \\ -\frac{14}{25} \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -\frac{7}{25} \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -\frac{7}{10} & 0 \\ -\frac{14}{25} & -\frac{7}{25} \end{bmatrix}, \qquad C = \begin{bmatrix} \frac{24}{25} & -\frac{7}{25} \\ \frac{7}{25} & \frac{24}{25} \end{bmatrix}.$$

Here $\det P = \frac{49}{250} \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$. All decimals in the question were treated as exact terminating rationals, so these entries are exact, not rounded numerical eigenpairs.

Exercise 5.5.20: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} -3 & -8 \\ 4 & 5 \end{bmatrix}.$$

The characteristic roots are $1 \pm (4)i$. Choose the root $\lambda = a - bi = 1 - 4i$ with $b = 4 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -8 \\ 4 - 4i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -8 \\ 4 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -4 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -8 & 0 \\ 4 & -4 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & -4 \\ 4 & 1 \end{bmatrix}.$$

Here $\det P = 32 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.21: Conjugation and real invariant directions

The first equation in Example 2 is $(-0.3 + 0.6i)x_1 - 0.6x_2 = 0$. Solve it for the second component:

$$x_2 = \frac{-0.3 + 0.6i}{0.6} x_1 = (-1/2 + i)x_1.$$

Taking $x_1 = 2$ gives $y = (2, -1 + 2i)^T$, as claimed. The original vector was $v_1 = (-2 - 4i, 5)^T$. With $c = (-1 + 2i)/5$, direct multiplication gives

$$c(-2 - 4i) = 2, \qquad c \cdot 5 = -1 + 2i.$$

Thus $y = cv_1$, with $c \neq 0$. They span the same complex eigenspace; different choices of a free parameter do not produce different eigenspace directions.

Exercise 5.5.22: Conjugation and real invariant directions

Suppose $Ax = \lambda x$ with $x \neq 0$ and choose a nonzero complex scalar μ. Complex linearity gives $A(\mu x) = \mu Ax = \mu\lambda x = \lambda(\mu x)$. Also $\mu x \neq 0$, because multiplying a zero result by $1/\mu$ would imply $x = 0$. Thus μx is an eigenvector for the same eigenvalue. Both its eigenvector equation and nonzero status are required.

Exercise 5.5.23: Conjugation and real invariant directions

Write $x^* = \overline{x}^T$ and $q = x^*Ax$, a 1×1 scalar. Here is the complete chain with the reasons for all five transitions:

$$\begin{aligned}
\overline{q} &= \overline{\overline{x}^T A x} \\
&= x^T \overline{A}\,\overline{x} && \text{(a): conjugation distributes over products;} \\
&= x^T A \overline{x} && \text{(b): } A \text{ has real entries;} \\
&= (x^T A \overline{x})^T && \text{(c): a scalar equals its transpose;} \\
&= \overline{x}^T A^T x && \text{(d): transpose reverses product order;} \\
&= \overline{x}^T A x = q && \text{(e): } A^T = A.
\end{aligned}$$

A complex number equals its conjugate exactly when its imaginary part is zero, so q is real. Conjugation does not reverse product order in (a); transposition does in (d).

Exercise 5.5.24: Conjugation and real invariant directions

Prove the eigenvalue is real without assuming a spectral theorem.

Let $x \neq 0$ and $Ax = \lambda x$. Multiply on the left by $x^* = \overline{x}^T$:

$$x^*Ax = \lambda x^*x, \qquad x^*x = \sum_{j=1}^{n} |x_j|^2 > 0.$$

The numerator is real by Exercise 23, so $\lambda = (x^*Ax)/(x^*x)$ is a real number.

Separate the real and imaginary components, checking nonzero status.

Write $x = u + iv$ with real u, v. Since A and λ are real, equating parts gives $Au = \lambda u$ and $Av = \lambda v$. At least one of u, v is nonzero because $x \neq 0$; every nonzero one is therefore a real eigenvector.

Conclusion. The eigenvalue is real, and $\operatorname{Re} x$ is an eigenvector *if it is nonzero*. The printed unconditional claim about the real part needs this qualification: for $A = I_2$ and $x = (i, 0)^T$, the real part is zero and is not an eigenvector. If the real part vanishes, the nonzero imaginary part supplies a real eigenvector instead; multiplying x by $-i$ makes that vector its real part.

Exercise 5.5.25: Conjugation and real invariant directions

Write $x = u + iv$ with $u = \operatorname{Re} x$ and $v = \operatorname{Im} x$. A real matrix times a real vector is real, and linearity gives $Ax = A(u + iv) = Au + iAv$. Uniqueness of the real and imaginary parts of each component

therefore yields

$$\operatorname{Re}(Ax) = A \operatorname{Re} x, \qquad \operatorname{Im}(Ax) = A \operatorname{Im} x.$$

The real-entry hypothesis matters: a matrix containing i could mix the real and imaginary parts.

Exercise 5.5.26: Conjugation and real invariant directions

(a) Expand the eigenvector equation before separating parts.

Let $v = u + iw$, with u, w real, and use the given $\lambda = a - bi$:

$$A(u + iw) = (a - bi)(u + iw) = (au + bw) + i(aw - bu).$$

Since A is real, comparison gives $Au = au + bw$ and $Aw = -bu + aw$, the required identities.

(b) Match the two columns of the matrix products.

With $P = [u\ w]$ and $C = \left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$,

$$AP = [Au\ Aw] = [au + bw\ \ -bu + aw] = PC.$$

For completeness, u, w are independent when $b \neq 0$: if v were a complex scalar multiple of one real vector, the eigenvector equation would force that nonzero real vector to be mapped by the real matrix to a nonreal scalar multiple of itself, impossible. Thus P is invertible, which is the additional fact needed to pass from $AP = PC$ to $A = PCP^{-1}$.

Exercise 5.5.27: Two real invariant planes from complex eigenpairs

What we must show. Produce an exact real block factorization of

$$A = \begin{bmatrix} 26 & 33 & 23 & 20 \\ -6 & -8 & -1 & -13 \\ -14 & -19 & -16 & 3 \\ -20 & -20 & -20 & -14 \end{bmatrix}.$$

The characteristic polynomial factors as $(t^2 + 4t + 29)\,(t^2 + 8t + 116)$. An exact eigenspace computation gives the following eigenvectors for one root in each conjugate pair:

$$\lambda = -2 - 5i, \qquad v = \begin{bmatrix} -1 - i \\ i \\ 1 \\ 0 \end{bmatrix}.$$

$$\lambda = -4 - 10i, \qquad v = \begin{bmatrix} -1 + i \\ -2i \\ 2i \\ 2 \end{bmatrix}.$$

For each pair choose the root with negative imaginary part. Write its eigenvector as $u + iw$, then put u, w into two adjacent columns of P. The identities in Exercise 26 supply the corresponding 2×2

block of C. This gives

$$P = \begin{bmatrix} -1 & -1 & -1 & 1 \\ 0 & 1 & 0 & -2 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 \end{bmatrix}.$$

$$C = \begin{bmatrix} -2 & -5 & 0 & 0 \\ 5 & -2 & 0 & 0 \\ 0 & 0 & -4 & -10 \\ 0 & 0 & 10 & -4 \end{bmatrix}.$$

The exact determinant is $\det P = 2 \neq 0$, and multiplying the displayed matrices gives $AP = PC$ with zero residual in every entry. Therefore $A = PCP^{-1}$. The calculation can be reproduced with exact SymPy arithmetic: factor `A.charpoly().as_expr()`, then for each selected root `lam` compute `(A-lam*eye(4)).nullspace()`. Extract componentwise real and imaginary parts for the two columns. Multiplying those columns by a common nonzero real scalar does not alter the block identity, which is why fractional vectors may be cleared to integers.

Exercise 5.5.28: Two real invariant planes from complex eigenpairs

What we must show. Produce an exact real block factorization of

$$A = \begin{bmatrix} 7 & 11 & 20 & 17 \\ -20 & -40 & -86 & -74 \\ 0 & -5 & -10 & -10 \\ 10 & 28 & 60 & 53 \end{bmatrix}.$$

The characteristic polynomial factors as $(t^2 - 6t + 10)\,(t^2 - 4t + 29)$. An exact eigenspace computation gives the following eigenvectors for one root in each conjugate pair:

$$\lambda = 3 - i, \qquad v = \begin{bmatrix} -2 \\ 2i \\ -3 - i \\ 4 \end{bmatrix}.$$

$$\lambda = 2 - 5i, \qquad v = \begin{bmatrix} 1 - i \\ -4 \\ 0 \\ 2 \end{bmatrix}.$$

For each pair choose the root with negative imaginary part. Write its eigenvector as $u + iw$, then put u, w into two adjacent columns of P. The identities in Exercise 26 supply the corresponding 2×2 block of C. This gives

$$P = \begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -4 & 0 \\ -3 & -1 & 0 & 0 \\ 4 & 0 & 2 & 0 \end{bmatrix}.$$

$$C = \begin{bmatrix} 3 & -1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 2 & -5 \\ 0 & 0 & 5 & 2 \end{bmatrix}.$$

The exact determinant is $\det P = -4 \neq 0$, and multiplying the displayed matrices gives $AP = PC$ with zero residual in every entry. Therefore $A = PCP^{-1}$. The calculation can be reproduced with exact SymPy arithmetic: factor `A.charpoly().as_expr()`, then for each selected root `lam` compute `(A-lam*eye(4)).nullspace()`. Extract componentwise real and imaginary parts for the two columns. Multiplying those columns by a common nonzero real scalar does not alter the block identity, which is why fractional vectors may be cleared to integers.

5.6 Discrete dynamical systems

For $x_{k+1} = Ax_k$, start by expanding x_0 in an eigenbasis: $x_0 = \sum_j c_j v_j$, $Av_j = \lambda_j v_j$. Then

$$x_k = \sum_j c_j \lambda_j^k v_j.$$

This follows by induction: multiplication by A adds one power of each eigenvalue. A mode with $|\lambda| < 1$ decays; one with $|\lambda| > 1$ grows. A negative eigenvalue also reverses its mode's sign at each step. The direction of fastest decay corresponds to the smallest modulus, whereas fastest growth corresponds to the largest modulus. These discrete-time tests concern modulus, not merely the sign of an eigenvalue.

Practice problem 5.6.P1: Determine the three initial coefficients

The given eigenvectors are $v_1 = (-2, 2, 1)^T$, $v_2 = (2, 1, 2)^T$, $v_3 = (1, 2, -2)^T$. Solve $c_1v_1 + c_2v_2 + c_3v_3 = (1, 11, -2)^T$:

$$-2c_1 + 2c_2 + c_3 = 1, \quad 2c_1 + c_2 + 2c_3 = 11, \quad c_1 + 2c_2 - 2c_3 = -2.$$

Elimination gives $(c_1, c_2, c_3) = (2, 1, 3)$; substitution checks all three coordinates. Therefore $x_k = 2v_1 + (2/3)^k v_2 + 3(1/3)^k v_3$.

Practice problem 5.6.P2: Retain only the nondecaying mode

The powers $(2/3)^k$ and $(1/3)^k$ tend to zero. Thus the preceding solution tends to $2v_1 = (-4, 4, 2)^T$, rather than zero. The eigenvalue 1 preserves its component indefinitely.

Exercise 5.6.1: Use the given eigenvectors without reconstructing the matrix

Solve $c_1(1, 1)^T + c_2(-1, 1)^T = (9, 1)^T$. The coordinate equations $c_1 - c_2 = 9$, $c_1 + c_2 = 1$ give $c_1 = 5, c_2 = -4$.

(a) Apply A once to the decomposition.

$x_1 = 5(3)(1, 1)^T - 4(1/3)(-1, 1)^T = (49/3, 41/3)^T$.

(b) Apply the same rule k times.

$$x_k = 5\,3^k(1, 1)^T - 4\,(1/3)^k(-1, 1)^T.$$

At $k = 0$ this gives the specified initial vector; multiplication by A verifies the recurrence at every subsequent step.

Exercise 5.6.2: Separate the growing mode from two decaying modes

The coefficient system $[v_1\ v_2\ v_3]c = x_0$ is

$$c_1 + 2c_2 - 3c_3 = -2, \quad c_2 - 3c_3 = -5, \quad -3c_1 - 5c_2 + 7c_3 = 3.$$

It gives $c = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$. Hence

$$x_k = 2 \cdot 3^k v_1 + (4/5)^k v_2 + 2(3/5)^k v_3.$$

The last two modes decay, whereas the first grows. More precisely, $x_k/3^k \to 2v_1 = (2, 0, -6)^T$. Thus the sequence is unbounded and its direction approaches the line through v_1; its second coordinate tends to zero while its first and third grow in opposite signs.

Exercise 5.6.3: Increase predation in the owl–wood-rat model

For $p = .2$ the matrix is $A = \begin{bmatrix} .5 & .4 \\ -.2 & 1.1 \end{bmatrix}$. The characteristic polynomial is $\lambda^2 - 1.6\lambda + .63$ with roots $.9, .7$. Eigenvectors are $(1, 1)^T$ and $(2, 1)^T$, as direct multiplication verifies. Thus

$$x_k = c_1(.9)^k(1, 1)^T + c_2(.7)^k(2, 1)^T.$$

Both modes decay. Under the exercise's $c_1 > 0$ assumption, the long-term positive populations both decline with factor approaching .9 per month. The first component counts owls; the second is the number of wood rats in thousands, as in Example 1.

Exercise 5.6.4: The borderline between eventual decline and growth

With $p = .125$, the matrix has eigenvalues $1, .6$ and eigenvectors $(4, 5)^T, (4, 1)^T$. For instance $A(4, 5)^T = (4, 5)^T$ and $A(4, 1)^T = .6(4, 1)^T$. Consequently

$$x_k = c_1(4, 5)^T + c_2(.6)^k(4, 1)^T \longrightarrow c_1(4, 5)^T.$$

The populations approach constant levels rather than tending to zero or growing indefinitely. Small parameter changes can move the largest eigenvalue below or above 1, changing the prediction to decline or growth. This explains the sensitivity called an unstable equilibrium in the exercise; the equilibrium line is not a robust attracting equilibrium under model changes. Coordinate ratio $4 : 5$ means 4 owls per 5 thousand rats under the model's units.

Exercise 5.6.5: Long-term growth and the dominant population ratio

For $p = .325$, $A = \begin{bmatrix} .4 & .3 \\ -.325 & 1.2 \end{bmatrix}$. Its eigenvalues solve $\lambda^2 - 1.6\lambda + .5775 = 0$, giving $1.05, .55$. A dominant eigenvector is $(6, 13)^T$, because $A(6, 13)^T = (6.3, 13.65)^T = 1.05(6, 13)^T$. Under the stated positive dominant-coefficient assumption, x_k is asymptotic to $c_1(1.05)^k(6, 13)^T$.

Conclusion. Both populations eventually grow about 5% per step, and their coordinate ratio tends to $6 : 13$. If the prey coordinate uses the thousands convention of Example 1, this is 6 owls per 13 thousand squirrels; coordinate scaling must be retained when interpreting a physical ratio.

Exercise 5.6.6: Choose the predation rate that preserves population levels

At $p = .5$, the characteristic polynomial is $\lambda^2 - 1.6\lambda + .63$, whose roots are $.9, .7$. Both powers tend to zero, so both populations perish in this model. For constant limiting levels, require an eigenvalue 1:

$$\det(A - I) = (-.6)(.2) - (.3)(-p) = -.12 + .3p = 0.$$

Thus $p = .4$. The other eigenvalue is .6, and a 1-eigenvector is $(1, 2)^T$. Therefore the solutions tend to $c_1(1, 2)^T$.

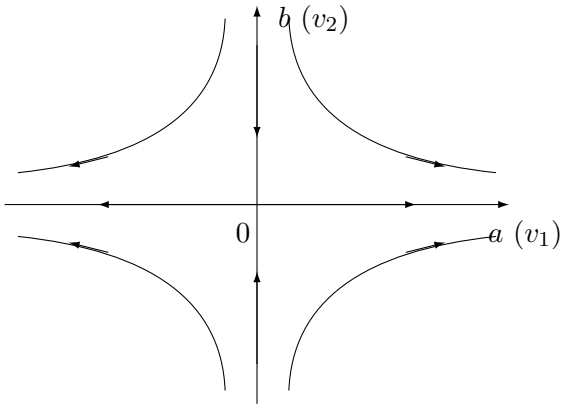
Conclusion. The preserving parameter is $p = .4$, and the limiting coordinate ratio is $1 : 2$, with prey-unit scaling interpreted as in the preceding exercise.

Exercise 5.6.7: A saddle with one attracting and one repelling direction

(a) Since $3 > 1$ but $1/3 < 1$, the origin is a saddle point.

(b) The line spanned by $v_1 = (1, 1)^T$ is repelling, with factor 3 each step. The line spanned by $v_2 = (-1, 1)^T$ is attracting, with factor $1/3$. These are also the greatest repulsion and attraction directions, respectively.

(c) In eigen-coordinates $x = av_1 + bv_2$, the update is $(a, b) \mapsto (3a, b/3)$. The sketch uses those coordinates: horizontal arrows point away from zero and vertical arrows toward zero. Off-axis trajectories approach the repelling axis while moving outward. The curves are schematic connections of discrete points, not continuous-time trajectories; ab remains constant here.



Exercise 5.6.8: Two stable modes and one unstable mode

The eigenvalues from Exercise 5.6.2 are $3, 4/5, 3/5$. A mode expands along $(1, 0, -3)^T$, while the two-dimensional span of the other eigenvectors is attracted to zero. The fastest contraction has the smaller modulus $3/5$, along $(-3, -3, 7)^T$; the greatest repulsion has modulus 3, along $(1, 0, -3)^T$.

Conclusion. The origin is a saddle. The initial vector in Exercise 2 has coefficient $c_1 = 2$, so its trajectory is repelled. Only initial vectors in the stable plane ($c_1 = 0$) tend to zero.

Exercise 5.6.9: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{17}{10} & -\frac{3}{10} \\ -\frac{6}{5} & \frac{4}{5} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(\lambda - 2)(2\lambda - 1)}{2} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{1}{2},\ v = \begin{bmatrix} 1 \\ 4 \end{bmatrix}; \qquad \lambda = 2,\ v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **saddle point**. The attracting direction is $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ (factor $\frac{1}{2}$), and the repelling direction is $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ (factor 2).

Exercise 5.6.10: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{3}{10} & \frac{2}{5} \\ -\frac{3}{10} & \frac{11}{10} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(2\lambda - 1)(10\lambda - 9)}{20} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{1}{2},\ v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{9}{10},\ v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **attractor**. The greatest attraction is along $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, for the smaller eigenvalue $\frac{1}{2}$. There is no repelling eigendirection.

Exercise 5.6.11: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{2}{5} & \frac{1}{2} \\ -\frac{2}{5} & \frac{13}{10} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(5\lambda - 4)(10\lambda - 9)}{50} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{4}{5},\ v = \begin{bmatrix} 5 \\ 4 \end{bmatrix}; \qquad \lambda = \frac{9}{10},\ v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **attractor**. The greatest attraction is along $\begin{bmatrix} 5 \\ 4 \end{bmatrix}$, for the smaller eigenvalue $\frac{4}{5}$. There is no repelling eigendirection.

Exercise 5.6.12: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{1}{2} & \frac{3}{5} \\ -\frac{3}{10} & \frac{7}{5} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(5\lambda - 4)(10\lambda - 11)}{50} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{4}{5},\ v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{11}{10},\ v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **saddle point**. The attracting direction is $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ (factor $\frac{4}{5}$), and the repelling direction is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (factor $\frac{11}{10}$).

Exercise 5.6.13: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{4}{5} & \frac{3}{10} \\ -\frac{2}{5} & \frac{3}{2} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(5\lambda - 6)(10\lambda - 11)}{50} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{11}{10},\ v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{6}{5},\ v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **repeller**. The greatest repulsion is along $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$, for the larger eigenvalue $\frac{6}{5}$. There is no attracting eigendirection.

Exercise 5.6.14: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{17}{10} & \frac{3}{5} \\ -\frac{2}{5} & \frac{7}{10} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(10\lambda - 13)(10\lambda - 11)}{100} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{11}{10},\ v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{13}{10},\ v = \begin{bmatrix} -3 \\ 2 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **repeller**. The greatest repulsion is along $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$, for the larger eigenvalue $\frac{13}{10}$. There is no attracting eigendirection.

Exercise 5.6.15: A stochastic model with a nonzero limiting vector

The columns of A sum to 1. The given $v_1 = (.1, .6, .3)^T$ has $Av_1 = v_1$. Eigenvectors for $.5, .2$ are $v_2 = (2, -3, 1)^T$ and $v_3 = (-1, 0, 1)^T$. Solving $c_1v_1 + c_2v_2 + c_3v_3 = (0, .3, .7)^T$ gives $c_1 = 1, c_2 = 1/10, c_3 = 3/10$. Thus

$$x_k = v_1 + \tfrac{1}{10}(.5)^k v_2 + \tfrac{3}{10}(.2)^k v_3.$$

At zero the coordinates are $(.1 + .2 - .3, .6 - .3, .3 + .1 + .3)$, as required. The last two modes tend to zero, so $x_k \to v_1$. The limiting proportions are $10\%, 60\%, 30\%$.

Exercise 5.6.16: The full Hertz three-location solution

In City Airport, Downtown, Metro Airport order, the matrix from Section 4.9 is

$$A = \frac{1}{100}\begin{bmatrix} 90 & 1 & 9 \\ 1 & 90 & 1 \\ 9 & 9 & 90 \end{bmatrix}.$$

Exact calculation gives three eigenpairs:

$$1,\ (91, 19, 99)^T; \qquad .89,\ (-1, 1, 0)^T; \qquad .81,\ (-1, 0, 1)^T.$$

The eigenvalues are distinct, so their eigenvectors form a basis. Therefore the general solution is

$$x_k = c_1(91, 19, 99)^T + c_2(.89)^k(-1, 1, 0)^T + c_3(.81)^k(-1, 0, 1)^T.$$

The constants are determined by the initial fleet distribution. For total fleet size N, adding the coordinates gives $N = 209c_1$, so the limit is $(N/209)(91, 19, 99)^T$. At $N = 2000$ the downtown limit is $2000(19)/209 \approx 181.818$ cars. This recovers the steady-state interpretation while providing the requested general dynamics.

Exercise 5.6.17: A two-stage growing population with all four requested graphs

(a) Translate births and survival into matrix entries.

Juveniles next year are born to this year's adults, while next year's adults are surviving juveniles plus surviving adults. Hence

$$A = \begin{bmatrix} 0 & 1.6 \\ .3 & .8 \end{bmatrix}.$$

(b) Find growth and the eventual ratio.

The characteristic polynomial is $\lambda^2 - .8\lambda - .48$, with roots $1.2, -.4$. A dominant eigenvector is $(4, 3)^T$; a second is $(-4, 1)^T$. The growing mode therefore has factor 1.2, a 20% annual growth rate, and limiting juvenile/adult ratio 4/3. For a nonzero nonnegative initial population the growing coefficient is positive: solving $x_0 = c_1(4, 3)^T + c_2(-4, 1)^T$ gives $c_1 = (j_0 + 4a_0)/16 > 0$.

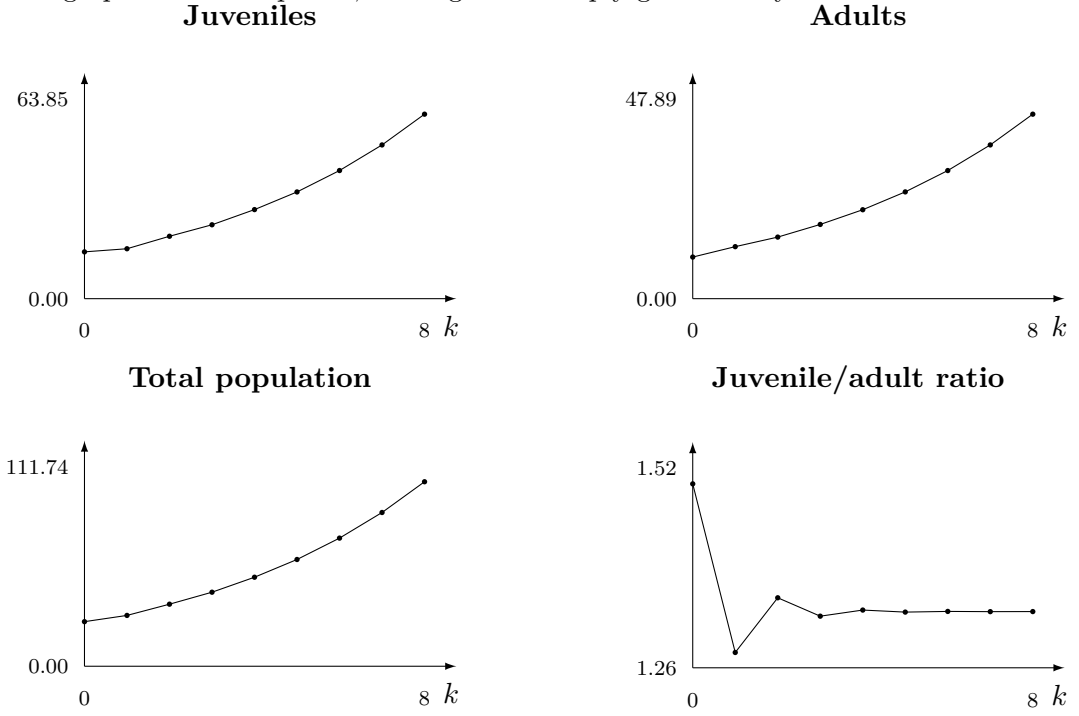
(c) Calculate the eight-year trajectory and plot it.

Beginning at $(15, 10)^T$, the actual iterates are

k	j_k	a_k	j_k/a_k
0	15.000000	10.000000	1.500000
1	16.000000	12.500000	1.280000
2	20.000000	14.800000	1.351351
3	23.680000	17.840000	1.327354
4	28.544000	21.376000	1.335329
5	34.201600	25.664000	1.332668
6	41.062400	30.791680	1.333555
7	49.266688	36.952064	1.333259
8	59.123302	44.341658	1.333358

.

The four graphs use these points; line segments simply guide the eye between successive integer years.



The ratio oscillates toward 4/3 because the secondary eigenvalue is negative. It is already close by years 3–4 and differs by less than .001 from 4/3 by year 5. The underlying relative mode shrinks by $|-.4/1.2| = 1/3$ each year. A complete plotting program, including the requested total and ratio panels, is:

```
import numpy as np
import matplotlib.pyplot as plt
A = np.array([[0.,1.6],[.3,.8]])
x = np.array([15.,10.])
values = [x.copy()]
for k in range(8):
    x = A @ x
    values.append(x.copy())
v = np.array(values)
series = [v[:,0],v[:,1],v.sum(axis=1),v[:,0]/v[:,1]]
```

```
labels = ['Juveniles','Adults','Total','Juveniles/adults']
fig, axes = plt.subplots(2,2)
for ax, y, label in zip(axes.flat, series, labels):
    ax.plot(range(9), y, 'o-')
    ax.set(xlabel='Year', ylabel=label)
fig.tight_layout()
plt.show()
```

Exercise 5.6.18: A three-stage buffalo model

(a) Construct the stage matrix.

Only adults give birth. Calves survive into the yearling stage and yearlings into adulthood. Thus

$$A = \begin{bmatrix} 0 & 0 & .42 \\ .60 & 0 & 0 \\ 0 & .75 & .95 \end{bmatrix}.$$

(b) Compute the dominant mode and its composition.

The characteristic polynomial is $\lambda^3 - .95\lambda^2 - .189$. Its real positive root is $\lambda_1 \approx 1.104834392$; the other two roots are complex conjugates of modulus less than 1. Since $\lambda_1 > 1$, the nonzero nonnegative population grows asymptotically by factor 1.104834392, about 10.483439% per year. Normalize the dominant eigenvector to 100 adults. The first row of $Av = \lambda_1 v$ gives $c = 42/\lambda_1$, and the second gives $y = .6c/\lambda_1$. Therefore the eventual composition per 100 adults is approximately

$$(c, y, a) = (38.014747, 20.644586, 100).$$

Substituting these formulas into the third row reproduces the characteristic equation; this checks the eigenvector normalization.

Show that every nonzero nonnegative initial population has a growing mode.

An initial state could in principle have zero coefficient in a particular eigenvector direction, so we must verify that the dominant coefficient is positive here. A useful row multiplier is the positive left eigenvector

$$w = \begin{bmatrix} .45/\lambda_1^2 \\ .75/\lambda_1 \\ 1 \end{bmatrix}.$$

Indeed direct multiplication gives

$$w^T A = \begin{bmatrix} .45/\lambda_1 & .75 & .189/\lambda_1^2 + .95 \end{bmatrix} = \lambda_1 w^T.$$

The last equality uses the characteristic equation, divided by λ_1^2: $\lambda_1 = .95 + .189/\lambda_1^2$. All three entries of w are strictly positive because $\lambda_1 > 0$.

The three distinct eigenvalues give an eigenvector expansion $x_0 = c_1 v_1 + c_2 v_2 + c_3 v_3$, allowing complex coefficients for the conjugate pair. For $j = 2, 3$, calculate the same scalar in two ways:

$$w^T A v_j = \lambda_1 w^T v_j = \lambda_j w^T v_j.$$

Since $\lambda_j \neq \lambda_1$, this forces $w^T v_j = 0$. Multiplying the expansion of x_0 by w^T therefore isolates

$$c_1 = \frac{w^T x_0}{w^T v_1} > 0.$$

The numerator is positive whenever x_0 has nonnegative entries and at least one positive entry; the denominator is positive because v_1 has the positive components computed above. Consequently

$$\frac{x_k}{\lambda_1^k} = c_1 v_1 + c_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k v_2 + c_3 \left(\frac{\lambda_3}{\lambda_1}\right)^k v_3 \longrightarrow c_1 v_1.$$

Both remaining ratios have modulus less than 1, so their powers tend to zero. Thus the dominant mode really is present for every nonzero nonnegative initial population, justifying both the growth rate and the eventual stage proportions claimed above.

5.7 Applications to differential equations

For continuous time the rule is $x'(t) = Ax(t)$. If $Av = \lambda v$, then $ve^{\lambda t}$ is a solution because its derivative is $\lambda ve^{\lambda t} = Ave^{\lambda t}$. For a real eigenbasis, expand the initial vector in that basis and replace each coefficient's evolution by $e^{\lambda t}$.

Complex eigenvalues. If $\lambda = a + bi$ and $v = p + iq$ with p, q real and $b > 0$, a conjugate pair of complex eigenfunctions is $ve^{(a+bi)t}$ and $\overline{v}e^{(a-bi)t}$. A real basis for the same solution space is

$$r_1(t) = e^{at}(p\cos bt - q\sin bt), \qquad r_2(t) = e^{at}(p\sin bt + q\cos bt).$$

This follows by multiplying out Euler's identity $e^{ibt} = \cos bt + i\sin bt$ and taking real and imaginary parts. For real initial data use arbitrary real coefficients of r_1, r_2. The sign of the real part a, rather than $|\lambda|$, decides continuous-time decay or growth.

Practice problem 5.7.P1: Diagonalize over the complex numbers

The three given eigenvalues $-.5, .2+.3i, .2-.3i$ are distinct. Their eigenvectors are consequently linearly independent over $\mathbb{C}$. The matrix $P = [v_1\ v_2\ v_3]$ is invertible and $A = P\,\mathrm{diag}(-.5, .2+.3i, .2-.3i)P^{-1}$.

Practice problem 5.7.P2: Convert the complex general solution to a real one

The complex general solution is $x = c_1v_1e^{-.5t} + c_2v_2e^{(.2+.3i)t} + c_3v_3e^{(.2-.3i)t}$, with complex constants. For real solutions use $v_1 = (1, -2, 1)^T$, $p = (1, 0, 2)^T$, and $q = (2, 4, 0)^T$. The real general solution is

$$x = d_1e^{-.5t}(1, -2, 1)^T + d_2r_1(t) + d_3r_2(t),$$

where r_1, r_2 are the formulas above with $a = .2, b = .3$ and all d_j real. This gives every real solution, since at zero the vectors v_1, p, q are linearly independent.

Practice problem 5.7.P3: Describe the stable line and unstable spiral plane

On the line spanned by v_1 trajectories approach zero like $e^{-.5t}$. In the plane spanned by p, q, they spiral outward with exponential factor $e^{.2t}$. A typical three-dimensional trajectory has both components: its stable-line component decays while its rotating planar component expands. The origin is a spiral saddle, not an attractor.

Exercise 5.7.1: Determine the constants from the initial position

Write $x(0) = c_1v_1 + c_2v_2$. In coordinates this is

$$\begin{bmatrix} -3 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -6 \\ 1 \end{bmatrix}.$$

Solving the two scalar equations gives $c_1 = \frac{5}{2}$, $c_2 = -\frac{3}{2}$. Therefore

$$x(t) = \frac{5}{2}e^{4t}\begin{bmatrix} -3 \\ 1 \end{bmatrix} + \left(-\frac{3}{2}\right)e^{2t}\begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

At $t = 0$ the coefficients reproduce the specified vector. Differentiating each term and using $Av_j = \lambda_jv_j$ verifies $x' = Ax$.

Exercise 5.7.2: Determine the constants from the initial position

Write $x(0) = c_1v_1 + c_2v_2$. In coordinates this is

$$\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Solving the two scalar equations gives $c_1 = \frac{1}{2}$, $c_2 = \frac{5}{2}$. Therefore

$$x(t) = \frac{1}{2}e^{-3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \left(\frac{5}{2}\right) e^{-1t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

At $t = 0$ the coefficients reproduce the specified vector. Differentiating each term and using $Av_j = \lambda_j v_j$ verifies $x' = Ax$.

Exercise 5.7.3: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda - 1)(\lambda + 1) = 0.$$

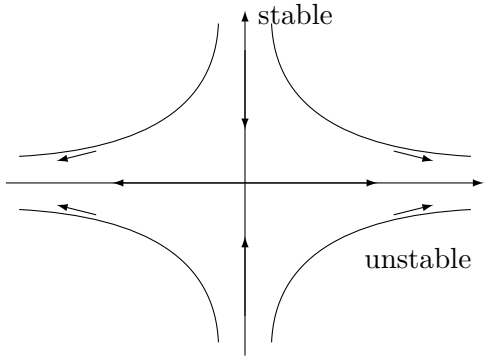
Its two eigenpairs are

$$\lambda_1 = -1,\ v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}; \quad \lambda_2 = 1,\ v_2 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}.$$

Solving $c_1v_1 + c_2v_2 = (3, 2)^T$ gives $(c_1, c_2) = (\frac{9}{2}, -\frac{5}{2})$. Hence the required initial-value solution is

$$x(t) = \left(\frac{9}{2}\right) e^{-1t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \left(-\frac{5}{2}\right) e^{1t} \begin{bmatrix} -3 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **saddle point**. The negative eigenvalue's eigenvector v_1 gives the attracting direction; the positive eigenvalue's v_2 gives the repelling direction. The schematic sketch is in these eigen-coordinates: vertical is stable, horizontal unstable. Transforming it by $[v_2\ v_1]$ gives the corresponding picture in the original plane.



Exercise 5.7.4: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} -2 & -5 \\ 1 & 4 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda - 3)(\lambda + 1) = 0.$$

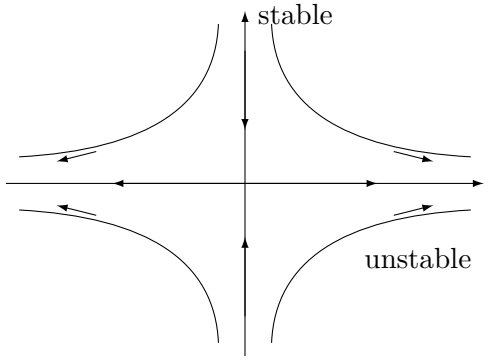
Its two eigenpairs are

$$\lambda_1 = -1,\ v_1 = \begin{bmatrix} -5 \\ 1 \end{bmatrix}; \quad \lambda_2 = 3,\ v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Solving $c_1 v_1 + c_2 v_2 = (3, 2)^T$ gives $(c_1, c_2) = (-\frac{5}{4}, \frac{13}{4})$. Hence the required initial-value solution is

$$x(t) = \left(-\frac{5}{4}\right) e^{-1t} \begin{bmatrix} -5 \\ 1 \end{bmatrix} + \left(\frac{13}{4}\right) e^{3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **saddle point**. The negative eigenvalue's eigenvector v_1 gives the attracting direction; the positive eigenvalue's v_2 gives the repelling direction. The schematic sketch is in these eigen-coordinates: vertical is stable, horizontal unstable. Transforming it by $[v_2\ v_1]$ gives the corresponding picture in the original plane.



Exercise 5.7.5: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} 7 & -1 \\ 3 & 3 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda - 6)(\lambda - 4) = 0.$$

Its two eigenpairs are

$$\lambda_1 = 4,\ v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}; \quad \lambda_2 = 6,\ v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Solving $c_1 v_1 + c_2 v_2 = (3, 2)^T$ gives $(c_1, c_2) = (-\frac{1}{2}, \frac{7}{2})$. Hence the required initial-value solution is

$$x(t) = \left(-\frac{1}{2}\right) e^{4t} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \left(\frac{7}{2}\right) e^{6t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **repeller**. Both exponential factors grow; the greater eigenvalue 6 gives the fastest repulsion, along $(1, 1)^T$. There is no attracting eigendirection.

Exercise 5.7.6: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda + 1)(\lambda + 2) = 0.$$

Its two eigenpairs are

$$\lambda_1 = -2,\ v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}; \quad \lambda_2 = -1,\ v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Solving $c_1 v_1 + c_2 v_2 = (3, 2)^T$ gives $(c_1, c_2) = (-1, 5)$. Hence the required initial-value solution is

$$x(t) = (-1)\, e^{-2t} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + (5)\, e^{-1t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **attractor**. Both exponents are negative. The fastest attraction is along $(2, 3)^T$, for eigenvalue -2; the other mode decays more slowly like e^{-t}. There is no repelling eigendirection.

Exercise 5.7.7: A constant eigenbasis decouples the differential equations

Choose the eigenvectors from Exercise 5.7.5 as columns:

$$P = \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 4 & 0 \\ 0 & 6 \end{bmatrix}.$$

Direct multiplication gives $AP = PD$, and $\det P = -2 \neq 0$. Put $x(t) = Py(t)$. Since P is constant, $x' = Py'$, so

$$Py' = APy = PDy, \qquad y' = Dy.$$

The last equality uses multiplication by P^{-1}. Thus $y_1' = 4y_1$ and $y_2' = 6y_2$, with no cross-coupling. Their scalar solutions are exponentials, and $x = Py$ converts back.

Exercise 5.7.8: A constant eigenbasis decouples the differential equations

Choose the eigenvectors from Exercise 5.7.6 as columns:

$$P = \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}.$$

Direct multiplication gives $AP = PD$, and $\det P = -1 \neq 0$. Put $x(t) = Py(t)$. Since P is constant, $x' = Py'$, so

$$Py' = APy = PDy, \qquad y' = Dy.$$

The last equality uses multiplication by P^{-1}. Thus $y_1' = -2y_1$ and $y_2' = -1y_2$, with no cross-coupling. Their scalar solutions are exponentials, and $x = Py$ converts back.

Exercise 5.7.9: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -3 & 2 \\ -1 & -1 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 4\lambda + 5.$$

A positive-imaginary eigenpair is

$$\lambda = -2 + i, \qquad v = \begin{bmatrix} 1 - i \\ 1 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(-2+i)t} + c_2 \overline{v} e^{(-2-i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{-2t}\big[p\cos(1t) - q\sin(1t)\big],$$
$$r_2 = e^{-2t}\big[p\sin(1t) + q\cos(1t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation in the original plane, and the negative real part produces decay. Every nonzero trajectory spirals inward to the origin.

Exercise 5.7.10: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 - 4\lambda + 5.$$

A positive-imaginary eigenpair is

$$\lambda = 2 + i, \qquad v = \begin{bmatrix} -1 - i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(2+i)t} + c_2 \overline{v} e^{(2-i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{2t}\big[p\cos(1t) - q\sin(1t)\big],$$
$$r_2 = e^{2t}\big[p\sin(1t) + q\cos(1t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation, while the positive real part expands its size. Nonzero trajectories spiral outward from the origin.

Exercise 5.7.11: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -3 & -9 \\ 2 & 3 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 9.$$

A positive-imaginary eigenpair is

$$\lambda = 3i, \qquad v = \begin{bmatrix} -3 + 3i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} -3 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(3i)t} + c_2 \overline{v} e^{(-3i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{0t}\big[p\cos(3t) - q\sin(3t)\big],$$
$$r_2 = e^{0t}\big[p\sin(3t) + q\cos(3t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is no exponential growth or decay. The real solutions trace closed ellipses around the origin, with period $2\pi/3$; the origin is a center.

Exercise 5.7.12: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -7 & 10 \\ -4 & 5 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 2\lambda + 5.$$

A positive-imaginary eigenpair is

$$\lambda = -1 + 2i, \qquad v = \begin{bmatrix} 3 - i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(-1+2i)t} + c_2 \overline{v} e^{(-1-2i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{-1t}\big[p\cos(2t) - q\sin(2t)\big],$$
$$r_2 = e^{-1t}\big[p\sin(2t) + q\cos(2t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation in the original plane, and the negative real part produces decay. Every nonzero trajectory spirals inward to the origin.

Exercise 5.7.13: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 4 & -3 \\ 6 & -2 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 - 2\lambda + 10.$$

A positive-imaginary eigenpair is

$$\lambda = 1 + 3i, \qquad v = \begin{bmatrix} 1+i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(1+3i)t} + c_2 \overline{v} e^{(1-3i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{1t}\big[p\cos(3t) - q\sin(3t)\big],$$
$$r_2 = e^{1t}\big[p\sin(3t) + q\cos(3t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation, while the positive real part expands its size. Nonzero trajectories spiral outward from the origin.

Exercise 5.7.14: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -2 & 1 \\ -8 & 2 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 4.$$

A positive-imaginary eigenpair is

$$\lambda = 2i, \qquad v = \begin{bmatrix} 1-i \\ 4 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(2i)t} + c_2 \overline{v} e^{(-2i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{0t}\big[p\cos(2t) - q\sin(2t)\big],$$
$$r_2 = e^{0t}\big[p\sin(2t) + q\cos(2t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is no exponential growth or decay. The real solutions trace closed ellipses around the origin, with period $2\pi/2$; the origin is a center.

Exercise 5.7.15: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -8 & -12 & -6 \\ 2 & 1 & 2 \\ 7 & 12 & 5 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda - 1)(\lambda + 1)(\lambda + 2).$$

All eigenvalues in this particular computer exercise are real, which is a special case of the requested complex-eigenfunction method. A complete eigenbasis is

$$\lambda_1 = -2, \quad v_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

$$\lambda_2 = -1, \quad v_2 = \begin{bmatrix} -6 \\ 1 \\ 5 \end{bmatrix}.$$

$$\lambda_3 = 1, \quad v_3 = \begin{bmatrix} -4 \\ 1 \\ 4 \end{bmatrix}.$$

Thus the general complex solution is

$$x(t) = \sum_{j=1}^{3} c_j e^{\lambda_j t} v_j, \qquad c_j \in \mathbb{C}.$$

The general real solution is the same formula with $c_j \in \mathbb{R}$, because this basis consists of real vectors. Each eigenpair checks by $Av_j = \lambda_j v_j$, and distinct eigenvalues give a full basis. There are two negative eigenvalues and one positive eigenvalue. The stable plane is $\mathrm{Span}\{v_1, v_2\}$ and the repelling line is $\mathrm{Span}\{v_3\}$. Typical trajectories first combine these modes and eventually move outward in the positive-eigenvalue direction: the origin is a saddle, with no oscillatory spiral in this example.

Exercise 5.7.16: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -6 & -11 & 16 \\ 2 & 5 & -4 \\ -4 & -5 & 10 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda - 4)(\lambda - 3)(\lambda - 2).$$

All eigenvalues in this particular computer exercise are real, which is a special case of the requested complex-eigenfunction method. A complete eigenbasis is

$$\lambda_1 = 2, \quad v_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

$$\lambda_2 = 3, \quad v_2 = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}.$$

$$\lambda_3 = 4, \quad v_3 = \begin{bmatrix} 7 \\ -2 \\ 3 \end{bmatrix}.$$

Thus the general complex solution is

$$x(t) = \sum_{j=1}^{3} c_j e^{\lambda_j t} v_j, \qquad c_j \in \mathbb{C}.$$

The general real solution is the same formula with $c_j \in \mathbb{R}$, because this basis consists of real vectors. Each eigenpair checks by $Av_j = \lambda_j v_j$, and distinct eigenvalues give a full basis. All three eigenvalues are positive. The origin is a repeller; generic trajectories eventually align with the fastest growing $\lambda = 4$ eigenvector, while exact invariant eigenspaces retain their own exponential behavior. There is no spiral because the spectrum is real.

Exercise 5.7.17: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 30 & 64 & 23 \\ -11 & -23 & -9 \\ 6 & 15 & 4 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda - 1)\left(\lambda^2 - 10\lambda + 29\right).$$

A positive-imaginary eigenpair is

$$\lambda = 5 + 2i, \qquad v = \begin{bmatrix} 23 - 34i \\ -9 + 14i \\ 3 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 23 \\ -9 \\ 3 \end{bmatrix}, \qquad q = \begin{bmatrix} -34 \\ 14 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(5+2i)t} + c_2 \overline{v} e^{(5-2i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{5t}\big[p\cos(2t) - q\sin(2t)\big],$$

$$r_2 = e^{5t}\big[p\sin(2t) + q\cos(2t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is also the real eigenpair $\lambda_0 = 1$, $w = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$. Add $c_0 e^{1t} w$ to the complex solution and $d_0 e^{1t} w$ to the real one. These three independent modes give the entire solution space. The real mode grows like e^t and the spiral-plane modes grow like e^{5t}. Thus the origin is repelling: generic trajectories spiral outward increasingly close to the invariant plane $\mathrm{Span}\{p, q\}$; only the real eigendirection has a nonrotating trajectory.

Exercise 5.7.18: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 53 & -30 & -2 \\ 90 & -52 & -3 \\ 20 & -10 & 2 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda + 7)\left(\lambda^2 - 10\lambda + 26\right).$$

A positive-imaginary eigenpair is

$$\lambda = 5 + i, \qquad v = \begin{bmatrix} 6 + 2i \\ 9 + 3i \\ 10 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 6 \\ 9 \\ 10 \end{bmatrix}, \qquad q = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(5+i)t} + c_2 \overline{v} e^{(5-i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{5t}\big[p\cos(1t) - q\sin(1t)\big],$$

$$r_2 = e^{5t}\big[p\sin(1t) + q\cos(1t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is also the real eigenpair $\lambda_0 = -7$, $w = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$. Add $c_0 e^{-7t} w$ to the complex solution and $d_0 e^{-7t} w$ to the real one. These three independent modes give the entire solution space. The line $\mathrm{Span}\{w\}$ is stable, decaying like e^{-7t}. The plane $\mathrm{Span}\{p, q\}$ spirals outward like e^{5t}. Thus this is a spiral saddle: only points on the stable line approach zero, and other trajectories eventually spiral outward.

Exercise 5.7.19: Substitute the circuit constants before solving the voltage system

Example 1 gives

$$A = \begin{bmatrix} -(1/R_1 + 1/R_2)/C_1 & 1/(R_2 C_1) \\ 1/(R_2 C_2) & -1/(R_2 C_2) \end{bmatrix} = \begin{bmatrix} -2 & \frac{3}{4} \\ 1 & -1 \end{bmatrix}.$$

Its eigenvalues are $-\frac{5}{2}, -\frac{1}{2}$, with eigenvector matrix $P = \begin{bmatrix} -3 & 1 \\ 2 & 2 \end{bmatrix}$. Solving $Pc = (4, 4)^T$ gives $c = \begin{bmatrix} -\frac{1}{2} \\ \frac{5}{2} \end{bmatrix}$. Therefore the requested voltage vector is

$$\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} = \left(-\frac{1}{2}\right) e^{-\frac{5}{2}t} \begin{bmatrix} -3 \\ 2 \end{bmatrix} + \left(\frac{5}{2}\right) e^{-\frac{1}{2}t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Each component at zero is the stipulated initial voltage, and differentiating verifies the given circuit equation. Both exponentials decay, consistent with a passive resistive circuit.

Exercise 5.7.20: Substitute the circuit constants before solving the voltage system

Example 1 gives

$$A=\begin{bmatrix}-(1/R_1+1/R_2)/C_1 & 1/(R_2C_1)\\ 1/(R_2C_2) & -1/(R_2C_2)\end{bmatrix}=\begin{bmatrix}-2 & \frac{1}{3}\\ \frac{3}{2} & -\frac{3}{2}\end{bmatrix}.$$

Its eigenvalues are $-\frac{5}{2},-1$, with eigenvector matrix $P=\begin{bmatrix}-2 & 1\\ 3 & 3\end{bmatrix}$. Solving $Pc=(3,3)^T$ gives $c=\begin{bmatrix}-\frac{2}{3}\\ \frac{5}{3}\end{bmatrix}$. Therefore the requested voltage vector is

$$\begin{bmatrix}v_1(t)\\ v_2(t)\end{bmatrix}=\left(-\frac{2}{3}\right)e^{-\frac{5}{2}t}\begin{bmatrix}-2\\ 3\end{bmatrix}+\left(\frac{5}{3}\right)e^{-1t}\begin{bmatrix}1\\ 3\end{bmatrix}.$$

Each component at zero is the stipulated initial voltage, and differentiating verifies the given circuit equation. Both exponentials decay, consistent with a passive resistive circuit.

Exercise 5.7.21: A damped oscillation in current and capacitor voltage

Using Example 3's circuit matrix with the specified constants gives

$$A=\begin{bmatrix}-R_2/L & -1/L\\ 1/C & -1/(R_1C)\end{bmatrix}=\begin{bmatrix}-1 & -8\\ 5 & -5\end{bmatrix}.$$

Its eigenvalues are $-3\pm 6i$. Put $B=A+3I$; direct multiplication gives $B^2=-36I$. Consequently the solution for $x_0=(0,15)^T$ is

$$x(t)=e^{-3t}\big[x_0\cos 6t+(Bx_0/6)\sin 6t\big].$$

Here $Bx_0=(-120,-30)^T$, so

$$i_L(t)=-20e^{-3t}\sin 6t,\qquad v_C(t)=e^{-3t}(15\cos 6t-5\sin 6t).$$

At zero these give 0 amps and 15 volts. Differentiating gives $x'=Ax$; the identity $B^2=-36I$ verifies the formula directly.

Exercise 5.7.22: Solve the second circuit with its stated sign convention

The problem supplies a different circuit matrix; with its constants,

$$A=\begin{bmatrix}0 & 1/L\\ -1/C & -1/(RC)\end{bmatrix}=\begin{bmatrix}0 & 2\\ -2/5 & -4/5\end{bmatrix}.$$

Its eigenvalues are $-2/5\pm(4/5)i$. Set $a=-2/5, b=4/5$ and $B=A-aI$. Then $B^2=-b^2I$ and $Bx_0=(24,-24/5)^T$ for $x_0=(0,12)^T$. The same cosine–sine construction gives

$$i_L(t)=30e^{-2t/5}\sin(4t/5),$$

$$v_C(t)=e^{-2t/5}\big[12\cos(4t/5)-6\sin(4t/5)\big].$$

The initial data and the differential equation both check directly. The signs differ from Exercise 21 because the supplied circuit and current-direction convention differ.

5.8 Iterative estimates for eigenvalues

Follow the book's indexing carefully. Starting with x_0, compute $y_k = Ax_k$, choose an entry μ_k of y_k with largest absolute value (retain its sign), and set $x_{k+1} = y_k/\mu_k$. Thus μ_1 comes from Ax_1, not Ax_0. A negative dominant eigenvalue need not cause the normalized vectors to alternate, because the signed normalization removes the reversal.

For inverse iteration with a shift α, solve $(A-\alpha I)y_k = x_k$ instead. Normalize by its largest-magnitude entry μ_k, and estimate the original eigenvalue by $\theta_k = \alpha + 1/\mu_k$. Reuse a factorization of $A - \alpha I$; do not explicitly form its inverse at every step. A small residual $\|Av - \theta v\|$ checks a proposed eigenpair. Agreement of two consecutive estimates alone can be misleading.

Practice problem 5.8.P1: Test proportionality and inspect the residual

The given vector $x = (1, -4.3, 8.1)^T$ gives $Ax = (3, -13, 24.5)^T$. The entrywise ratios are 3, $130/43 \approx 3.023256$, and $245/81 \approx 3.024691$. Their close agreement suggests an eigenvalue near 3.024. The Rayleigh estimate is

$$R(x) = \frac{x^T Ax}{x^T x} = \frac{25735}{8510} \approx 3.024089.$$

For $\theta = R(x)$, inspect $Ax - \theta x$ relative to the size of Ax or x. A residual near zero, rather than the raw size of x, is the meaningful approximation test. Exact proportionality would give zero residual and an exact eigenvector.

Exercise 5.8.1: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} 1 \\ \frac{1663}{5000} \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} \frac{24989}{5000} \\ \frac{4163}{2500} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx 4.997800$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} 1 \\ \frac{1}{3} \end{bmatrix}$, and direct multiplication checks $Av = 5v$ exactly.

Conclusion. The dominant eigenvalue is 5, with eigenvector $\begin{bmatrix} 1 \\ \frac{1}{3} \end{bmatrix}$.

Exercise 5.8.2: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} -\frac{63}{250} \\ 1 \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} -\frac{1567}{1250} \\ \frac{3129}{625} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx 5.006400$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} -\frac{1}{4} \\ 1 \end{bmatrix}$, and direct multiplication checks $Av = 5v$ exactly.

Conclusion. The dominant eigenvalue is 5, with eigenvector $\begin{bmatrix} -\frac{1}{4} \\ 1 \end{bmatrix}$.

Exercise 5.8.3: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} \frac{1297}{2500} \\ 1 \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} \frac{2297}{5000} \\ \frac{2836}{3125} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx 0.907520$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$, and direct multiplication checks $Av = \frac{9}{10}v$ exactly.

Conclusion. The dominant eigenvalue is $\frac{9}{10}$, with eigenvector $\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$.

Exercise 5.8.4: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} 1 \\ \frac{3751}{5000} \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} -\frac{1003}{2500} \\ -\frac{3761}{12500} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx -0.401200$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} 1 \\ \frac{3}{4} \end{bmatrix}$, and direct multiplication checks $Av = -\frac{2}{5}v$ exactly.

Conclusion. The dominant eigenvalue is $-\frac{2}{5}$, with eigenvector $\begin{bmatrix} 1 \\ \frac{3}{4} \end{bmatrix}$.

Here "largest" in the exercise means largest in *absolute value*: the other eigenvalue is .1, which is algebraically larger but not dominant.

Exercise 5.8.5: Normalize by the second component of the supplied iterate

Dividing the last vector by its second entry gives

$$x = \begin{bmatrix} -\frac{24991}{31241} \\ 1 \end{bmatrix} \approx \begin{bmatrix} -\frac{7999}{10000} \\ 1 \end{bmatrix}.$$

Use the requested four-decimal vector for a check:

$$A \begin{bmatrix} -\frac{7999}{10000} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{8003}{2000} \\ -\frac{2501}{500} \end{bmatrix}.$$

The second entry estimates the eigenvalue by $-\frac{2501}{500}$, since the corresponding input entry is 1. The nearby exact eigenpair is $\lambda = -5$, $v = \begin{bmatrix} -\frac{4}{5} \\ 1 \end{bmatrix}$, verified by direct multiplication. The rounded estimate is approximate, so exact equality $Ax = \lambda x$ is not expected until using the exact vector.

Exercise 5.8.6: Normalize by the second component of the supplied iterate

Dividing the last vector by its second entry gives

$$x = \begin{bmatrix} -\frac{2045}{4093} \\ 1 \end{bmatrix} \approx \begin{bmatrix} -\frac{1249}{2500} \\ 1 \end{bmatrix}.$$

Use the requested four-decimal vector for a check:

$$A \begin{bmatrix} -\frac{1249}{2500} \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{2501}{1250} \\ \frac{5003}{1250} \end{bmatrix}.$$

The second entry estimates the eigenvalue by $\frac{5003}{1250}$, since the corresponding input entry is 1. The nearby exact eigenpair is $\lambda = 4$, $v = \begin{bmatrix} -\frac{1}{2} \\ 1 \end{bmatrix}$, verified by direct multiplication. The rounded estimate is approximate, so exact equality $Ax = \lambda x$ is not expected until using the exact vector.

Exercise 5.8.7: Compute the requested five indexed power-method rows

Use the signed largest-magnitude entry for normalization and do not round intermediate vectors. For $A = \begin{bmatrix} 6 & 7 \\ 8 & 5 \end{bmatrix}$ and $x_0 = (1, 0)^T$, the results are

k	$(x_k)_1$	$(x_k)_2$	μ_k
1	0.750000000	1.000000000	11.500000000
2	1.000000000	0.956521739	12.782608696
3	0.993197279	1.000000000	12.959183673
4	1.000000000	0.998950131	12.994750656
5	0.999838416	1.000000000	12.999030499

.

For an indexing check, the first multiplication gives $Ax_0 = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$ and $\mu_0 = 8$. Dividing by this scalar produces x_1, after which μ_1 is computed from Ax_1. The values converge toward the dominant eigenvalue 13, with eigenvector direction $(1, 1)^T$.

Exercise 5.8.8: Compute the requested five indexed power-method rows

Use the signed largest-magnitude entry for normalization and do not round intermediate vectors. For $A = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$ and $x_0 = (1, 0)^T$, the results are

k	$(x_k)_1$	$(x_k)_2$	μ_k
1	0.500000000	1.000000000	7.000000000
2	0.285714286	1.000000000	6.142857143
3	0.255813953	1.000000000	6.023255814
4	0.250965251	1.000000000	6.003861004
5	0.250160772	1.000000000	6.000643087

.

For an indexing check, the first multiplication gives $Ax_0 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ and $\mu_0 = 4$. Dividing by this scalar produces x_1, after which μ_1 is computed from Ax_1. The values converge toward the dominant eigenvalue 6, with eigenvector direction $(1/4, 1)^T$.

Exercise 5.8.9: Continue the same indexed algorithm to the sixth estimate

For the given matrix

$$A = \begin{bmatrix} 8 & 0 & 12 \\ 1 & -2 & 1 \\ 0 & 3 & 0 \end{bmatrix}, \qquad x_0 = (1, 0, 0)^T,$$

repeatedly compute Ax_k, retain its signed largest-magnitude entry, and divide to form x_{k+1}. This gives

$$\mu_5 \approx 8.4233109850, \qquad \mu_6 \approx 8.4246301920.$$

The accompanying current vectors, which make these estimates reproducible, are

$$x_5 \approx \begin{bmatrix} 1.0 \\ 0.0993553378 \\ 0.0352759154 \end{bmatrix}, \qquad x_6 \approx \begin{bmatrix} 1.0 \\ 0.0993154879 \\ 0.0353858493 \end{bmatrix}.$$

Multiplying each vector by A and selecting its largest-magnitude entry checks the listed scalars. The vectors have been rounded only for display, so the check agrees to the displayed precision.

Exercise 5.8.10: Continue the same indexed algorithm to the sixth estimate

For the given matrix

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 9 \\ 0 & 1 & 9 \end{bmatrix}, \qquad x_0 = (1, 0, 0)^T,$$

repeatedly compute Ax_k, retain its signed largest-magnitude entry, and divide to form x_{k+1}. This gives

$$\mu_5 \approx 9.9319371728, \qquad \mu_6 \approx 9.9871727289.$$

The accompanying current vectors, which make these estimates reproducible, are

$$x_5 \approx \begin{bmatrix} 0.0183246073 \\ 1.0 \\ 0.9904013962 \end{bmatrix}, \qquad x_6 \approx \begin{bmatrix} 0.0037778949 \\ 1.0 \\ 0.9981549815 \end{bmatrix}.$$

Multiplying each vector by A and selecting its largest-magnitude entry checks the listed scalars. The vectors have been rounded only for display, so the check agrees to the displayed precision.

Exercise 5.8.11: Compare the scaling estimate with the Rayleigh quotient

The exact dominant eigenvalue, checked from the characteristic polynomial, is 6. Compute each Rayleigh quotient from the current x_k using $R(x_k) = x_k^T A x_k / (x_k^T x_k)$:

k	μ_k	$R(x_k)$	$\lvert\mu_k - \lambda\rvert$	$\lvert R(x_k) - \lambda\rvert$
1	5.800000000000	5.965517241379	$2.000e-01$	$3.448e-02$
2	5.965517241379	5.999035679846	$3.448e-02$	$9.643e-04$
3	5.994219653179	5.999973208305	$5.780e-03$	$2.679e-05$
4	5.999035679846	5.999999255782	$9.643e-04$	$7.442e-07$

.

The quotient's error is substantially smaller at the same iterate. Both matrices are symmetric, so this demonstrates the improved accuracy claimed in the exercise. Signed normalization is especially important for the negative dominant eigenvalue in Exercise 12.

Exercise 5.8.12: Compare the scaling estimate with the Rayleigh quotient

The exact dominant eigenvalue, checked from the characteristic polynomial, is -4. Compute each Rayleigh quotient from the current x_k using $R(x_k) = x_k^T A x_k / (x_k^T x_k)$:

k	μ_k	$R(x_k)$	$\lvert\mu_k - \lambda\rvert$	$\lvert R(x_k) - \lambda\rvert$
1	-4.333333333333	-3.923076923077	$3.333e-01$	$7.692e-02$
2	-3.923076923077	-3.995121951220	$7.692e-02$	$4.878e-03$
3	-4.019607843137	-3.999694842844	$1.961e-02$	$3.052e-04$
4	-3.995121951220	-3.999980926586	$4.878e-03$	$1.907e-05$

.

The quotient's error is substantially smaller at the same iterate. Both matrices are symmetric, so this demonstrates the improved accuracy claimed in the exercise. Signed normalization is especially important for the negative dominant eigenvalue in Exercise 12.

Exercise 5.8.13: A theoretically convergent method can be impractically slow

If the two leading eigenvalues have different moduli, there is a strictly dominant one, so the power method converges for an initial vector with a nonzero component in that eigendirection (under the section's eigenbasis hypotheses). But the ratio of the other large modulus to the dominant modulus is near $4/4 = 1$. The unwanted mode therefore decays slowly relative to the desired one.

Conclusion. It can work, but may require many iterations and be less useful than shifted inverse iteration. An initial vector with no dominant component is an exception to the usual convergence claim.

Exercise 5.8.14: Break the tie by shifting toward the positive eigenvalue

If the leading eigenvalues are exactly λ and $-\lambda$, their powers have equal moduli; ordinary power iteration need not select a single direction. Choose a shift α near the positive eigenvalue (for example 4.1, if suitably close) but not exactly an eigenvalue. Inverse iteration applies to $(A-\alpha I)^{-1}$. Its eigenvalues are $1/(\lambda_j-\alpha)$, so the desired positive eigenvalue is dominant when its distance from the shift is uniquely smallest. Solving shifted systems and normalizing then gives a sequence for that positive eigenpair. Alternatively, adding a suitable positive multiple of I can create a unique positive dominant modulus; subtract that shift from the resulting eigenvalue estimate. The inverse method is usually faster when an accurate initial estimate is available.

Exercise 5.8.15: Transform an eigenvalue under a shifted inverse

From $Ax=\lambda x$ subtract αx to get $(A-\alpha I)x=(\lambda-\alpha)x$. Since α is not an eigenvalue, the matrix is invertible and $\lambda-\alpha\neq 0$. Multiply by $(A-\alpha I)^{-1}$:

$$x=(\lambda-\alpha)(A-\alpha I)^{-1}x, \qquad (A-\alpha I)^{-1}x=\frac{1}{\lambda-\alpha}x.$$

Because $x\neq 0$, this is exactly the claimed eigenpair.

Exercise 5.8.16: Recover the original eigenvalue from the inverse one

Multiply $(A-\alpha I)^{-1}x=\mu x$ by $A-\alpha I$: $x=\mu(A-\alpha I)x$. The inverse is invertible, so its eigenvalue μ is nonzero. Dividing by μ and rearranging gives

$$Ax=\left(\alpha+\frac{1}{\mu}\right)x.$$

Thus $\alpha+1/\mu$ is an eigenvalue of A, with the same nonzero eigenvector. This proves the scalar update in inverse iteration.

Exercise 5.8.17: Run shifted inverse iteration and verify its residual

Use

$$A=\begin{bmatrix}10&-8&-4\\-8&13&4\\-4&5&4\end{bmatrix}, \quad \alpha=3.3, \quad x_0=(1,0,0)^T.$$

At each step solve $(A-\alpha I)y_k=x_k$, choose a signed largest-magnitude component μ_k, and set $x_{k+1}=y_k/\mu_k$, $\theta_k=\alpha+1/\mu_k$. Using high-precision arithmetic, iteration index $k=7$ gives

$$\theta_k\approx 3.3212201246571, \qquad x_{k+1}\approx\begin{bmatrix}1.0\\0.787012129501\\0.0956707098337\end{bmatrix}.$$

Before rounding, $\|Ax_{k+1}-\theta_k x_{k+1}\|_\infty<10^{-10}$ and the last two scalar estimates differ by less than 10^{-12}. The eigenvalue to four decimal places is $\boxed{3.3212}$. Independent roots of the exact characteristic polynomial confirm this choice. The shift makes this eigenvalue uniquely closest, so inverse iteration selects it rather than the dominant eigenvalue of A.

Exercise 5.8.18: Run shifted inverse iteration and verify its residual

Use

$$A = \begin{bmatrix} 8 & 0 & 12 \\ 1 & -2 & 1 \\ 0 & 3 & 0 \end{bmatrix}, \quad \alpha = -1.4, \quad x_0 = (1, 0, 0)^T.$$

At each step solve $(A - \alpha I)y_k = x_k$, choose a signed largest-magnitude component μ_k, and set $x_{k+1} = y_k/\mu_k$, $\theta_k = \alpha + 1/\mu_k$. Using high-precision arithmetic, iteration index $k = 10$ gives

$$\theta_k \approx -1.424428900898, \qquad x_{k+1} \approx \begin{bmatrix} 1.0 \\ 0.372900802803 \\ -0.785369075075 \end{bmatrix}.$$

Before rounding, $\|Ax_{k+1} - \theta_k x_{k+1}\|_\infty < 10^{-10}$ and the last two scalar estimates differ by less than 10^{-12}. The eigenvalue to four decimal places is $\boxed{-1.4244}$. Independent roots of the exact characteristic polynomial confirm this choice. The shift makes this eigenvalue uniquely closest, so inverse iteration selects it rather than the dominant eigenvalue of A.

Exercise 5.8.19: Compute both extreme-modulus eigenpairs from the required start

Use $x_0 = (1, 0, 0, 0)^T$ for both computations.

(a) Ordinary power iteration.

With signed largest-component normalization, the scalar estimate converges to

$$\lambda_{\max} \approx 30.288685345802, \quad v_{\max} \approx \begin{bmatrix} 0.95762878047 \\ 0.68893692056 \\ 1.0 \\ 0.94378150874 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{30.2887}$.

(b) Inverse iteration with shift zero.

Repeatedly solve $Ay_k = x_k$ instead of explicitly multiplying by A^{-1}. The smallest-modulus eigenpair is

$$\lambda_{\text{near }0} \approx 0.010150048397894, \quad v_{\text{near }0} \approx \begin{bmatrix} -0.60397234234 \\ 1.0 \\ -0.25113513051 \\ 0.14895344556 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{0.0102}$. The stopping tests were both a scalar-estimate change below 10^{-12} and an unrounded residual below 10^{-10}. They hold at indices $k = 16$ and $k = 6$, respectively, with $k = 0$ representing the first multiplication or solve. Independent characteristic-polynomial roots verify which eigenvalues are largest and nearest zero. The displayed eigenvectors are scaled to make their largest-magnitude component equal to 1.

Exercise 5.8.20: Compute both extreme-modulus eigenpairs from the required start

Use $x_0 = (1, 0, 0, 0)^T$ for both computations.

(a) Ordinary power iteration.

With signed largest-component normalization, the scalar estimate converges to

$$\lambda_{\max} \approx 19.182036763332, \quad v_{\max} \approx \begin{bmatrix} 0.1844163575 \\ 1.0 \\ 0.17961525893 \\ 0.40710960748 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{19.1820}$.

(b) Inverse iteration with shift zero.

Repeatedly solve $Ay_k = x_k$ instead of explicitly multiplying by A^{-1}. The smallest-modulus eigenpair is

$$\lambda_{\text{near } 0} \approx 0.012205562828844, \quad v_{\text{near } 0} \approx \begin{bmatrix} 1.0 \\ 0.22260978915 \\ -0.91799334087 \\ 0.66048300357 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{0.0122}$. The stopping tests were both a scalar-estimate change below 10^{-12} and an unrounded residual below 10^{-10}. They hold at indices $k = 18$ and $k = 6$, respectively, with $k = 0$ representing the first multiplication or solve. Independent characteristic-polynomial roots verify which eigenvalues are largest and nearest zero. The displayed eigenvectors are scaled to make their largest-magnitude component equal to 1.

Exercise 5.8.21: Separate vector size from convergence of direction

For $x = (.5, .5)^T$, diagonal multiplication is explicit:

(a) $A^k x = (.5(.8)^k, .5(.2)^k)^T \to 0$. The zero vector is not an eigenvector. Yet the second coordinate relative to the first is $(1/4)^k \to 0$, so the direction tends to the first coordinate axis.

(b) $A^k x = (.5, .5(.8)^k)^T \to (.5, 0)^T$, a nonzero eigenvector. Here the dominant eigenvalue is exactly 1, so its scale is preserved.

(c) $A^k x = (.5(8)^k, .5(2)^k)^T$ grows without bound. Its direction again tends to the first coordinate axis, but it does not converge to a finite vector.

The common conclusion is about normalized direction, not convergence of $A^k x$ itself. Moreover, a vector exactly missing the dominant mode cannot reveal that eigendirection. Appropriate normalization is therefore an essential part of the power method, not an optional cosmetic step.

Reproducing the iterative computations. The following complete functions implement the signed indexing used in the tables. Enter the matrix printed in the chosen exercise and its stated initial vector as NumPy arrays. For 19–20 and inverse iteration, continue until both the estimate change and eigenpair residual are small.

```
import numpy as np
def power_rows(A, x, count):
    for k in range(count+1):
        y = A @ x
        mu = y[np.argmax(np.abs(y))]
        rayleigh = (x @ A @ x)/(x @ x)
        yield k, x.copy(), mu, rayleigh
        x = y/mu

def inverse_rows(A, alpha, x, count):
    H = A-alpha*np.eye(len(x))
    for k in range(count+1):
        y = np.linalg.solve(H, x)
        mu = y[np.argmax(np.abs(y))]
        xnew = y/mu
        theta = alpha+1/mu
        residual = np.max(np.abs(A@xnew-theta*xnew))
        yield k, xnew.copy(), theta, residual
        x = xnew
```

All numerical results above were actually computed from the supplied matrices; the code is provided to reproduce them. Extra working precision was used for the final residual checks in Exercises 17–20.

5.S Chapter 5 supplementary exercises

These exercises connect eigenvectors, similarity, polynomial expressions in matrices, and diagonalization. Keep three distinctions in view: row equivalence does not preserve eigenvalues; similarity does. An eigenvalue may be zero, but an eigenvector must be nonzero. A repeated eigenvalue does not by itself decide diagonalizability: the total number of independent eigenvectors decides it. All matrices are real unless complex roots or complex diagonalization are explicitly being considered.

Exercise 5.S.1: Twenty-four statements with proofs or counterexamples

(a) **True.** If $Ax = x$ and $x \neq 0$, multiplication by A^{-1} gives $A^{-1}x = x$.

(b) **False.** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is invertible and row equivalent to I, but has only one independent eigenvector, since its 1-eigenspace is $\text{Span}\{(1,0)^T\}$.

(c) **True.** A zero column makes $Ae_j = 0$ for some nonzero coordinate vector. A zero row makes the matrix singular as well; by the Invertible Matrix Theorem its kernel contains a nonzero vector.

(d) **False.** The scalar matrix [2] has eigenvalue 2, whereas its square [4] has only eigenvalue 4.

(e) **True.** $Ax = \lambda x$ implies $A^2x = \lambda^2 x$, with the same nonzero x.

(f) **True.** An invertible A has $\lambda \neq 0$; $Ax = \lambda x$ gives $A^{-1}x = \lambda^{-1}x$.

(g) **False.** The zero matrix has eigenvalue zero.

(h) **True.** Nonzeroness is part of the eigenvector definition; otherwise $A0 = \lambda 0$ would make every scalar an eigenvalue.

(i) **False.** For I_2, the independent vectors e_1, e_2 have the same eigenvalue 1.

(j) **True.** Similarity gives $\det(P^{-1}(A - \lambda I)P) = \det(A - \lambda I)$, so the characteristic polynomials, including multiplicities, agree.

(k) **False.** For $A = \text{diag}(1, 2)$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ the eigenvalues agree and the matrices are similar (both have independent eigenbases), but e_2 is an eigenvector of A and not of B.

(l) **False.** For $A = \text{diag}(1, 2)$, $e_1 + e_2$ is not an eigenvector. Even within one eigenspace, the sum of v and $-v$ is zero and hence not an eigenvector.

(m) **False.** All diagonal entries of an upper triangular matrix are eigenvalues, including zeros.

(n) **True.** $\det(A^T - \lambda I) = \det((A - \lambda I)^T) = \det(A - \lambda I)$.

(o) **False.** I_5 has only one distinct eigenvalue but is already diagonal; its five coordinate vectors form an eigenbasis.

(p) **True.** The real rotation matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues $\pm i$ and no real eigenvectors.

(q) **False.** $\text{diag}(0, 1)$ is diagonalizable but its first column is zero, so its columns are dependent.

(r) **True.** If $Av = \lambda v = \mu v$ and $v \neq 0$, then $(\lambda - \mu)v = 0$ forces $\lambda = \mu$.

(s) **False.** Invertibility and diagonalizability are different. The non-diagonalizable invertible example in (b) refutes one direction; $\operatorname{diag}(0, 1)$ refutes the reverse direction.

(t) **True.** If $Ae_j = \lambda_j e_j$, column j has only its diagonal entry possibly nonzero, so A is diagonal.

(u) **True.** If $A = SBS^{-1}$ and $B = PDP^{-1}$, then $A = (SP)D(SP)^{-1}$, a diagonalization.

(v) **True.** $A^{-1}(AB)A = BA$, so the products are similar. In fact invertibility of A alone suffices.

(w) **False.** The zero matrix has n independent eigenvectors but is not invertible.

(x) **True.** Diagonalizability supplies an eigenvector basis, and the definition of a basis says every vector is a linear combination of its vectors (over $\mathbb{R}$ for real diagonalization here).

Exercise 5.S.2: Move an eigenvector through the right-hand factor

Suppose $(AB)x = \lambda x$ and $Bx \neq 0$. Associativity gives

$$(BA)(Bx) = B(ABx) = B(\lambda x) = \lambda(Bx).$$

Because Bx is nonzero by assumption, it is an eigenvector of BA for the same eigenvalue. The condition $Bx \neq 0$ is what prevents the zero vector from being incorrectly called an eigenvector.

Exercise 5.S.3: Apply two polynomials to an eigenvector

(a) $(5I - A)x = 5x - Ax = (5 - \lambda)x$. Thus x remains a nonzero eigenvector, with eigenvalue $5 - \lambda$.

(b) First $A^2x = A(\lambda x) = \lambda^2 x$. Consequently $(5I - 3A + A^2)x = (5 - 3\lambda + \lambda^2)x$. The new eigenvalue is $5 - 3\lambda + \lambda^2$.

Exercise 5.S.4: Induct on the matrix power

For $m = 1$ the assertion is the assumed eigenvector equation. Suppose $A^m x = \lambda^m x$. Multiplying by A gives $A^{m+1}x = A(\lambda^m x) = \lambda^m Ax = \lambda^{m+1}x$. The same x remains nonzero at every stage, so each displayed scalar is genuinely an eigenvalue of the corresponding power. This proves the claim for every positive integer m.

Exercise 5.S.5: Polynomial evaluation preserves eigenvectors

Exercise 5.S.4 gives $A^j x = \lambda^j x$ for every positive power; for $j = 0$ the same equality is $Ix = x$. Therefore

$$p(A)x = \sum_{j=0}^{n} c_j A^j x = \sum_{j=0}^{n} c_j \lambda^j x = p(\lambda)x.$$

Since $x \neq 0$, $p(\lambda)$ is an eigenvalue of $p(A)$. Different eigenvalues of A can map to the same polynomial value; that does not invalidate the conclusion.

Exercise 5.S.6: Diagonalize polynomial expressions using the original basis

(a) With $A = PDP^{-1}$, neighboring factors $P^{-1}P$ cancel in A^2. Since $I = PIP^{-1}$, $5I - 3A + A^2 = P(5I - 3D + D^2)P^{-1}$. At diagonal entries $2, 7$, the polynomial has values $3, 33$. Thus $B = P\,\mathrm{diag}(3, 33)P^{-1}$ is a diagonalization.

(b) The same cancellation gives $A^j = PD^jP^{-1}$ for every power. Adding the finitely many terms gives $p(A) = Pp(D)P^{-1}$, with diagonal entries $p(2), p(7)$. This remains a diagonalization even if those two values coincide.

Exercise 5.S.7: Cayley–Hamilton for a diagonalizable matrix

Write $A = PDP^{-1}$. Every diagonal entry λ_j of D is an eigenvalue of A and hence a root of its characteristic polynomial p. By the polynomial identity in Exercise 5.S.6,

$$p(A) = Pp(D)P^{-1} = P\,\mathrm{diag}(p(\lambda_1), \ldots, p(\lambda_n))P^{-1} = P0P^{-1} = 0.$$

This proves the theorem under the stated diagonalizability assumption; it does not pretend that every square matrix is already known to be diagonalizable.

Exercise 5.S.8: One eigenvalue plus diagonalizability forces a scalar matrix

(a) If the only eigenvalue is λ with multiplicity n, a diagonalization has $D = \lambda I$. Therefore $A = P(\lambda I)P^{-1} = \lambda PP^{-1} = \lambda I$.

(b) The given $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ has only eigenvalue 3, twice, but is not $3I$. By (a), it cannot be diagonalizable. Directly, $(A - 3I)x = 0$ forces $x_2 = 0$, leaving only one independent eigenvector.

Exercise 5.S.9: An eigenvalue inside the unit disk cannot equal one

If $I - A$ were singular, a nonzero v would satisfy $(I - A)v = 0$. Then $Av = v$, so 1 would be an eigenvalue of A. Its modulus is 1, contradicting the strict bound on every eigenvalue. Hence $I - A$ is invertible. This argument requires no diagonalizability assumption.

Exercise 5.S.10: Every eigenmode decays and therefore the whole matrix does

Use $A = PDP^{-1}$. Its powers satisfy $A^k = PD^kP^{-1}$. Every diagonal entry of D^k is $\lambda_j^k \to 0$ because $|\lambda_j| < 1$. Thus each entry of PD^kP^{-1} is a finite sum of fixed constants times quantities tending to zero, so it also tends to zero. Equivalently, apply the mode decomposition to each coordinate vector e_j; then $A^k e_j \to 0$, which says each column of A^k tends to zero.

Conclusion. $A^k \to 0$ entry by entry.

Exercise 5.S.11: Invariant lines and real eigenvectors

(a) Every $x \in H$ has the form $x = cu$. Then $Ax = cAu = c\lambda u \in H$. This is exactly invariance.

(b) Choose a nonzero u spanning the one-dimensional invariant subspace K. Invariance gives $Au \in K$, so $Au = \lambda u$ for some scalar λ. Since $u \neq 0$, it is an eigenvector.

Exercise 5.S.12: Factor a block triangular characteristic polynomial

In the permutation expansion of $G = \begin{bmatrix} A & X \\ 0 & B \end{bmatrix}$, a nonzero term must assign the bottom block rows to bottom block columns; otherwise it uses a zero from the lower-left block. The remaining top rows use top block columns. The products and signs therefore separate into the determinant sums for A and B, giving $\det G = \det A \det B$. The same argument applied to $G - \lambda I = \begin{bmatrix} A - \lambda I & X \\ 0 & B - \lambda I \end{bmatrix}$ gives the product of the two characteristic polynomials. No inverse of either diagonal block is required.

Exercise 5.S.13: Read one eigenvalue and solve the remaining block

The upper-left block is $[3]$ and the lower-right block is $B = \begin{bmatrix} 5 & -2 \\ -4 & 3 \end{bmatrix}$. Its characteristic polynomial is $(\lambda - 5)(\lambda - 3) - 8 = \lambda^2 - 8\lambda + 7 = (\lambda - 1)(\lambda - 7)$. By Exercise 5.S.12 the complete eigenvalue list is $3, 1, 7$, each with multiplicity one. The entries in the upper-right block have no effect on this list.

Exercise 5.S.14: Combine the spectra of two diagonal blocks

The blocks are $B = \begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} -7 & -4 \\ 3 & 1 \end{bmatrix}$. Their characteristic polynomials are

$$\det(\lambda I - B) = \lambda^2 - 5\lambda - 6 = (\lambda - 6)(\lambda + 1),$$

$$\det(\lambda I - C) = \lambda^2 + 6\lambda + 5 = (\lambda + 1)(\lambda + 5).$$

Multiply them by the block result. The eigenvalues are $6, -5$, each once, and -1 twice. Algebraic multiplicity is counted even when an eigenvalue occurs in both blocks.

Exercise 5.S.15: Split into the constant vector and coordinate-sum-zero space

Let $\mathbf{1} = (1, \ldots, 1)^T$. Then $J\mathbf{1} = n\mathbf{1}$, so $A\mathbf{1} = [a + (n-1)b]\mathbf{1}$. For z with coordinate sum zero, $Jz = 0$, and hence $Az = (a - b)z$. The coordinate-sum-zero subspace has dimension $n - 1$; together with $\mathbf{1}$ it gives a basis of $\mathbb{R}^n$. Thus the characteristic polynomial is

$$(\lambda - (a - b))^{n-1}(\lambda - (a + (n-1)b)).$$

When $b \neq 0$ and $n \geq 2$, the two eigenvalues are distinct, with multiplicities $n - 1$ and 1. If $b = 0$, they coincide at a and the combined multiplicity is n; for $n = 1$ only the value a is present. These qualifications prevent double-counting a coincident value.

Exercise 5.S.16: Apply the common-diagonal formula twice

For the 3×3 matrix, $a = 1, b = 2$. Thus $a - b = -1$ has multiplicity 2, and $a + 2b = 5$ has multiplicity 1. For the 5×5 matrix, $a = 7, b = 3$. Thus $a - b = 4$ has multiplicity 4, and $a + 4b = 19$ has multiplicity 1. These multiplicities add to the dimension in each case.

Exercise 5.S.17: The trace and determinant decide reality in dimension two

For a real matrix $A = [a_{ij}]$, expand

$$\det(\lambda I - A) = (\lambda - a_{11})(\lambda - a_{22}) - a_{12}a_{21} = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$

A real quadratic has both roots real exactly when its discriminant is nonnegative. Here that condition is $(\operatorname{tr} A)^2 - 4 \det A \geq 0$, equivalently

$$\det A \leq \left(\frac{\operatorname{tr} A}{2}\right)^2.$$

Equality includes a repeated real eigenvalue; it does not by itself assert diagonalizability.

Exercise 5.S.18: Compute the limiting spectral projection

The matrix has eigenvalues 1 and $3/5$, with eigenvectors $(-1, 2)^T$ and $(-3, 2)^T$. Distinct eigenvalues give a basis. The projection onto the 1-eigenspace along the other eigenspace is

$$E = \frac{A - (3/5)I}{1 - 3/5} = \begin{bmatrix} -1/2 & -3/4 \\ 1 & 3/2 \end{bmatrix}.$$

Indeed E acts as identity on the first eigenvector and as zero on the second, so on the whole eigenbasis

$$A^k = E + (3/5)^k(I - E).$$

The second term tends to zero. Therefore A^k tends to the stated matrix E, whose negative entries cause no contradiction: the original A already has a negative off-diagonal entry.

Exercise 5.S.19: A quadratic companion matrix

For $p(t) = 6 - 5t + t^2$, the companion matrix in the book's convention is

$$C_p = \begin{bmatrix} 0 & 1 \\ -6 & 5 \end{bmatrix}.$$

Its characteristic polynomial is $\det(\lambda I - C_p) = \det \begin{bmatrix} \lambda & -1 \\ 6 & \lambda - 5 \end{bmatrix} = \lambda^2 - 5\lambda + 6 = p(\lambda)$. If using $\det(C_p - \lambda I)$ instead, the sign is the same in even dimension.

Exercise 5.S.20: A cubic companion matrix and its determinant

The coefficients $(-24, 26, -9, 1)$ give

$$C_p = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 24 & -26 & 9 \end{bmatrix}.$$

Expand $\det(\lambda I - C_p)$ along the first row:

$$\lambda \det \begin{bmatrix} \lambda & -1 \\ 26 & \lambda - 9 \end{bmatrix} - 24 = \lambda^3 - 9\lambda^2 + 26\lambda - 24.$$

This equals $(\lambda - 2)(\lambda - 3)(\lambda - 4) = p(\lambda)$. For the book's $\det(C_p - \lambda I)$ convention the polynomial is its negative, as the dimension is odd.

Exercise 5.S.21: Inductively compute the companion characteristic polynomial

For $n = 2$, direct expansion gives $\det \begin{bmatrix} -\lambda & 1 \\ -a_0 & -a_1 - \lambda \end{bmatrix} = \lambda^2 + a_1\lambda + a_0 = (-1)^2 p(\lambda)$. Suppose the identity holds in dimension $n - 1$. Expand $C_p - \lambda I$ down its first column. Only the top and bottom entries can contribute. The top contribution is $-\lambda$ times the determinant of the companion-form block associated with $q(t) = a_1 + a_2 t + \cdots + a_{n-1}t^{n-2} + t^{n-1}$. By induction this determinant is $(-1)^{n-1}q(\lambda)$. The bottom contribution is $(-a_0)(-1)^{n+1}$ times a determinant of 1; the latter minor is triangular with diagonal ones. Thus the total is

$$(-\lambda)(-1)^{n-1}q(\lambda) + (-1)^n a_0 = (-1)^n[\lambda q(\lambda) + a_0] = (-1)^n p(\lambda).$$

This completes the induction for all $n \geq 2$.

Exercise 5.S.22: A polynomial root supplies an explicit companion eigenvector

(a) The companion matrix is $C_p = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix}$.

(b) Since $p(\lambda) = 0$, $\lambda^3 = -a_0 - a_1\lambda - a_2\lambda^2$. Therefore

$$C_p(1, \lambda, \lambda^2)^T = (\lambda, \lambda^2, -a_0 - a_1\lambda - a_2\lambda^2)^T = (\lambda, \lambda^2, \lambda^3)^T = \lambda(1, \lambda, \lambda^2)^T.$$

The first coordinate is 1, so the vector is nonzero even if $\lambda = 0$; it is genuinely an eigenvector.

Exercise 5.S.23: A Vandermonde matrix is an eigenvector matrix

Each column of V is the eigenvector from Exercise 5.S.22 for one root λ_j. Distinct roots give distinct eigenvalues, so the corresponding eigenvectors are independent. Thus V is invertible without needing to compute its inverse. The three eigenvector equations combine into $C_p V = V \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$. Multiply on the left by V^{-1} to obtain the requested diagonal matrix. Complex roots are allowed if matrices and independence are interpreted over $\mathbb{C}$.

Exercise 5.S.24: The companion-matrix idea behind a polynomial-root command

The basic method documented for MATLAB's `roots` is to construct a companion matrix and compute its eigenvalues. After removing leading zero coefficients and normalizing the leading nonzero coefficient, the polynomial is monic. Exercise 5.S.21 shows that the companion matrix's eigenvalues are exactly its roots. Numerical eigenvalue algorithms, typically using QR-based methods, then approximate those values. The companion matrix used by a program may be transposed or written in a different convention; its characteristic polynomial still supplies the same roots. The MATLAB reference is https://www.mathworks.com/help/matlab/ref/roots.html. This explains the algorithmic idea, not an assertion that every computed root has exact arithmetic accuracy: companion matrices can be sensitive to coefficient rounding, especially near repeated roots.

Exercise 5.S.25: A repeated eigenvalue prevents an exact diagonalization

For the supplied matrix, exact symbolic computation gives

$$\det(\lambda I - A) = (\lambda - 1)(\lambda - 2)^2.$$

Its eigenspaces are

$$E_1 = \mathrm{Span}\{(1, -2, 2)^T\}, \qquad E_2 = \mathrm{Span}\{(2, -5, 3)^T\}.$$

Each has dimension one, so there are only two independent eigenvectors for a three-dimensional matrix. Exact diagonalization is impossible. To see what happens to the requested checks, choose

$$D = \mathrm{diag}(1, 2, 2), \quad P = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -5 & -5 \\ 2 & 3 & 3 \end{bmatrix}.$$

Then $AP - PD = 0$, but $\det P = 0$ because columns 2 and 3 coincide. Thus PDP^{-1} is undefined. A zero eigenvector-equation residual alone does not make P an invertible change of basis. Floating-point eigenvalue commands may split the repeated root slightly and return nearly parallel eigenvectors. An attempted inverse then amplifies rounding errors; exact ranks resolve the issue. The following executed symbolic checks reproduce the conclusion:

```
from sympy import Matrix, eye, symbols, factor
A = Matrix([[-3,-2,0],[14,7,-1],[-6,-3,1]])
z = symbols('z')
print(factor(A.charpoly(z).as_expr()))
print((A-eye(3)).nullspace())
print((A-2*eye(3)).nullspace())
P = Matrix([[1,2,2],[-2,-5,-5],[2,3,3]])
D = Matrix.diag(1,2,2)
assert A*P-P*D == Matrix.zeros(3)
assert P.det() == 0
```

Exercise 5.S.26: A nonzero nilpotent matrix cannot be diagonalized

For the given 4×4 matrix, exact calculation gives

$$\det(\lambda I - A) = \lambda^4, \qquad \mathrm{Nul}\, A = \mathrm{Span}\{(1, 2, 1, 0)^T\}.$$

The only eigenvalue is zero, with algebraic multiplicity four but only one independent eigenvector. Thus the matrix is not diagonalizable. In fact $A^4 = 0$ and $A^3 \neq 0$, another exact check of its nonzero nilpotent structure. An eigenvector matrix for $D = 0_4$ can only have columns proportional to $(1, 2, 1, 0)^T$. Such a P has rank at most one: $AP - PD = 0$ is possible, but P^{-1} is impossible. Were an invertible P to exist, $PDP^{-1} = 0$ would contradict the given nonzero A. A floating-point eigenvalue routine can again produce small nonzero numbers around the repeated zero root and unreliable eigenvectors. The exact calculation that settles the issue is:

```
A = Matrix([[-8,5,-2,0],[-5,2,1,-2],
            [10,-8,6,-3],[3,-2,1,0]])
assert A.charpoly().as_expr().expand() == A.charpoly().gen**4
assert len(A.nullspace()) == 1
assert A**4 == Matrix.zeros(4)
assert A**3 != Matrix.zeros(4)
```

No external data are needed, and the failure is a proved mathematical obstruction, not a software or setup failure.